

Comparative morphology of gigantopyramidal neurons in primary motor cortex across mammals

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Abstract

Gigantopyramidal neurons, referred to as Betz cells in primates, are characterized by large somata and extensive basilar dendrites. Although there have been morphological descriptions and drawings of gigantopyramidal neurons in a limited number of species, quantitative investigations have typically been limited to measures of soma size. The current study thus employed two separate analytical approaches: a morphological investigation using the Golgi technique to provide qualitative and quantitative somatodendritic measures of gigantopyramidal neurons across 19 mammalian species from 7 orders; and unbiased stereology to compare the soma volume of layer V pyramidal and gigantopyramidal neurons in primary motor cortex between 11 carnivore and 9 primate species. Of the 617 neurons traced in the morphological analysis, 181 were gigantopyramidal neurons, with deep (primarily layer V) pyramidal ($n = 203$) and superficial (primarily layer III) pyramidal ($n = 233$) neurons quantified for comparative purposes. Qualitatively, dendritic morphology varied considerably across species, with some (sub)orders (e.g., artiodactyls, perissodactyls, feliforms) exhibiting bifurcating, V-shaped apical dendrites. Basilar dendrites exhibited idiosyncratic geometry across and within taxonomic groups. Quantitatively, most dendritic measures were significantly greater in gigantopyramidal neurons than in superficial and deep pyramidal neurons. Cluster analyses revealed that most taxonomic groups could be discriminated based on somatodendritic morphology for both superficial and gigantopyramidal neurons. Finally, in agreement with

Brodmann, gigantopyramidal neurons in both the morphological and stereological analyses were larger in feliforms (especially in the *Panthera* species) than in other (sub)orders, possibly due to specializations in muscle fiber composition and musculoskeletal systems.

KEYWORDS

brain evolution, dendrite, Golgi method, morphometry, neocortex, stereology, RRID:nif-0000-10294

1 | INTRODUCTION

Gigantopyramidal neurons in primary motor cortex (M1), referred to as Betz cells in primates (Betz, 1874), are generally characterized by large somata (Brodmann, 1909) and idiosyncratic basilar dendritic arrays (Scheibel & Scheibel, 1978a; Walshe, 1942). Investigations of these neurons have historically focused on soma size and shape (Vogt & Vogt, 1942; von Bonin, 1938; von Economo & Koskinas, 1925), with only qualitative descriptions of dendritic morphology in a limited number of species: human (Braak & Braak, 1976; Rivara, Sherwood, Bouras, & Hof, 2003), monkey (species unspecified; Gatter, Sloper, & Powell, 1978), domestic cat (Kaiserman-Abramof & Peters, 1972; Lewis, 1878), and sheep (Ebinger, 1975; Lewis, 1878). Betz initially observed similarities in the form and location of these “Riesenpyramiden” (p. 578) between primates and canids. Later, Brodmann observed variation in gigantopyramidal soma size across species. Recently, quantitative investigations have included measures of dendritic extent in individual species (giraffe: Jacobs, Harland, et al., 2015; Siberian tiger, clouded leopard: Johnson et al., 2016). Nevertheless, there are currently no quantitative studies comparing dendritic measures in gigantopyramidal neurons across multiple species. To this end, using available tissue of sufficient quality, the present investigation documents both qualitative and quantitative aspects of gigantopyramidal neurons in the primary motor cortices of 19 species across 7 phylogenetic orders: carnivores: suborder caniforms (African wild dog, domestic dog), carnivores: suborder feliforms (banded mongoose, caracal, clouded leopard, Siberian tiger, African lion), perissodactyls (mountain zebra, plains zebra), artiodactyls (blue wildebeest, greater kudu, giraffe), primates (ring-tailed lemur, golden lion tamarin, chacma baboon, human), a lagomorph (Flemish giant rabbit), a murid rodent (Long-Evans rat), and a diprotodont marsupial (Bennett’s wallaby). To supplement these findings and to determine whether Brodmann’s observations of substantially larger gigantopyramidal neurons in carnivores than in primates hold across a broader range of species, we also employed unbiased stereology to explore layer V pyramidal and gigantopyramidal neuron volumes in M1 between carnivores (11 species) and primates (9 species).

In humans, early counts suggested approximately 30,000 Betz cells per hemisphere (Campbell, 1905; Lassek, 1940; Scheibel & Scheibel, 1978a); however, more recent stereological investigation suggests over four times this number (~125,290) in layer Vb of left M1 (Rivara et al., 2003). Within M1, 75% of these neurons are located in the dorsal third of the gyrus, which contains representations of the lower limb, 18% are found in the medial third, corresponding to the upper limb, and 7%

are located in the inferior third, representing the face region (Lassek, 1940). Both soma size and density are graded along the mediolateral axis of the precentral gyrus such that the largest somata are found in areas controlling the lower limb, with the greatest density of gigantopyramidal neurons located in areas controlling the upper limb (Lassek, 1940; Rivara et al., 2003; von Bonin, 1949). Physiologically, gigantopyramidal neurons appear to be phasically active, exhibiting a rapid bursting pattern believed to introduce partial inhibition of the extensor muscles along with flexor facilitation prior to a movement, and a restoration of extensor tone immediately after the movement (Evarts, 1965, 1967; Lundberg & Voorhoeve, 1962). It has therefore been suggested that these neurons modulate the output of surrounding deep pyramidal neurons during the initiation of motor programs (Scheibel, Davies, Lindsay, & Scheibel, 1974). For the lower limb, this system appears to play a major role in the control of anti-gravity muscles involved in posture and locomotion (Scheibel & Scheibel, 1978a; Scheibel et al., 1974; Scheibel, Tomiyasu, & Scheibel, 1977). For the upper limb, gigantopyramidal neurons appear to be involved in the fine motor control of the hand and wrist (Lemon, 2008; Lemon, Kirkwood, Maier, Nakajima, & Nathan, 2004). Apart from these physiological findings in primates and the domestic cat, the specific functional contribution of gigantopyramidal neurons to motor control remains largely unexplored.

Insofar as gigantopyramidal neurons exhibit considerable variation in soma size, shape, and distribution, they may not constitute a distinct neuronal subtype (Braak & Braak, 1976; Rivara et al., 2003; Walshe, 1942). Nevertheless, they can generally be characterized by their somatodendritic characteristics. Gigantopyramidal neuron soma size in primates can be up to 20 times greater than in typical pyramidal neurons (Rivara et al., 2003), and they may have up to 15 basilar dendrites (Betz, 1874; Scheibel et al., 1974; Sherwood et al., 2003) compared to the 4–7 found in typical pyramidal neurons (Jacobs, Driscoll, & Schall, 1997; Jacobs et al., 2001; Scheibel & Scheibel, 1978a). Basilar dendrites of gigantopyramidal neurons can exhibit idiosyncratic arrangements, exiting the soma circumferentially or asymmetrically, extending up to 3 mm in any direction, occasionally resulting in long, obliquely descending taproots (Hammer, Tomiyasu, & Scheibel, 1979; Scheibel & Scheibel, 1978a; Scheibel et al., 1977). Additionally, in contrast to rodent and primate pyramidal neurons, which tend to have singularly ascending apical dendrites (Feldman, 1984; Parnavelas, Lieberman, & Webster, 1977; Peters & Walsh, 1972), the apical dendrites of gigantopyramidal neurons often fork near the soma into daughter branches of similar diameter (Deschênes, Labelle, & Landry, 1979; Kaiserman-Abramof & Peters, 1972).

From a comparative perspective, gigantopyramidal neurons in carnivores exhibit the largest somata (Brodman, 1909), which tend to be wide and rounded in shape (Betz, 1874; Johnson et al., 2016). In primates, somata tend to be relatively large (Brodman, 1909; Sherwood et al., 2003) and conically shaped (Braak & Braak, 1976). In artiodactyls, somata are more intermediate in size and tend to be elongated in shape (Badlangana, Bhagwandin, Fuxe, & Manger, 2007; Ebinger, 1975; Lewis, 1878). Marsupials and rodents exhibit the smallest gigantopyramidal somata, although it remains unclear if they exist in these species as a separate neuronal subtype (Brodman, 1909). For those species that have been examined quantitatively, the number of primary basilar dendritic segments for gigantopyramidal neurons appears to be much larger in the giraffe (~13; Jacobs, Harland, et al., 2015) than in the tiger and clouded leopard (~3; Johnson et al., 2016). These limited observations suggest that potential morphological differences may extend across species. Thus, the current study consists of two separate analyses: a morphological investigation using the Golgi technique to provide qualitative and quantitative somatodendritic measures of gigantopyramidal neurons (and of superficial and deep pyramidal neurons for comparative purposes) across mammalian species; and an unbiased stereological approach to document the soma volume of layer V pyramidal and gigantopyramidal neurons in M1 between carnivores and primates. Qualitatively, we predicted that the somata of feliform gigantopyramidal neurons would be wider and more rounded than those of other (sub)orders. Additionally, based on previous research (Barasa, 1960; Butti et al., 2015; Johnson et al., 2016) we expected the morphological analysis to reveal that gigantopyramidal neurons in perissodactyls, artiodactyls, and feliforms would tend to be characterized by V-shaped apical dendrite bifurcations. Quantitatively, in part because soma size and dendritic extent tend to be positively correlated (Jacobs et al., 1997, 2001), we expected both analyses to reveal that gigantopyramidal neurons would be larger in caniforms, and especially feliforms, than in other (sub)orders. Finally, based on our previous comparative research in a variety of species (Jacobs et al., 2014; Jacobs, Harland, et al., 2015; Johnson et al., 2016), we expected that the constellation of somatodendritic measures could be used to statistically differentiate the taxonomic groups.

2 | MATERIALS AND METHODS

Methodological procedures for the neuromorphological investigation are presented first, followed by those for the stereological analysis.

2.1 | Neuronal morphology

2.1.1 | Specimens

M1 samples were obtained from 19 species (23 animals) across 7 orders that were available for morphological analysis: 7 carnivores (2 caniforms: African wild dog, domestic dog; 5 feliforms: banded mongoose, caracal, clouded leopard, Siberian tiger, African lion); 2 perissodactyls (mountain zebra, plains zebra), 3 artiodactyls (2 blue wildebeests, greater kudu, 2 giraffes), 4 primates (ring-tailed lemur,

golden lion tamarin, chacma baboon, human), a lagomorph (Flemish giant rabbit), a murid rodent (2 Long-Evans rats), and a diprotodont (Bennett's wallaby). The phylogenetic relationships among these species are illustrated in Figure 1. Background details of each specimen in the morphological analysis are provided in Table 1. Antemortem observations for captive animals revealed no obvious neural deficits or behavioral abnormalities; observations of animals obtained from their natural habitat were not possible. Postmortem examination revealed no gross neuroanatomical abnormalities. The present study was approved by the Colorado College Institutional Review Board (#011311-1) and the University of the Witwatersrand Animal Ethics Committee (#2012/53/1), which complies with NIH policies for the care and use of animals in scientific experimentation.

2.1.2 | Tissue selection

Tissue blocks (3–5 mm-thick) were extracted from M1 of each species, except for the rat where intact coronal blocks were sectioned due to the small size of the brain. Tissue was removed based on the documented location of M1 across several species: domestic cat (Kaiserman-Abramof & Peters, 1972; Scheibel et al., 1974), mongoose (Radinsky, 1975), giraffe (Badlangana et al., 2007), horse, ox, dog (Hof, Bogaert, Rosenthal, & Fiskum, 1996; Levine, Levine, Hoffman, & Bratton, 2008), baboon (Waters, Samulack, Dykes, & McKinley, 1990), tamarin monkey, lemur (Sherwood et al., 2003), human (Jacobs et al., 2001; Rivara et al., 2003), rat (Paxinos & Watson, 1986), and wallaby (Lende, 1963). M1 pyramidal neurons recently documented with the same methodology in three species (giraffe: Jacobs, Harland, et al., 2015; clouded leopard and Siberian tiger: Johnson et al., 2016) were also included in the present sample to provide a broader comparative framework—exact locations of M1 tissue blocks are provided in these articles. In all species for the present study, the goal was to remove tissue midway through the “homunculus,” in a dorsolateral position approximating the representation of the hand-forepaw region (Boroojerdi et al., 1999; Rivara et al., 2003; Yousry et al., 1997). In primates, there is a high likelihood that this dorsolateral sampling location corresponds to the hand-forepaw region based on available somatotopic mapping in this phylogenetic order. A block of tissue, oriented rostrocaudally but parallel to the pial surface, spanning the primary somatosensory and motor cortices was removed and stained. In primates and carnivores, we confirmed this sampling location through histological identification of M1, among other areas including 3b and 3a (Figure 2). The border between 3a and M1 was identified, with M1 lacking a layer IV and having a large number of gigantopyramidal neurons. For other orders, although the equivalent area was more difficult to define, gigantopyramidal neurons could be identified with relative confidence in all but four species (i.e., mongoose, rabbit, rat, and wallaby) in the morphological analysis, indicating that sampled regions represented M1. Prior to tracing, tissue was coded to minimize experimenter bias, stained with a modified rapid Golgi technique (Scheibel & Scheibel, 1978b), and serially sectioned at 120 μ m with a vibratome (Leica VT1000S, Leica Microsystems, Wetzlar, Germany).

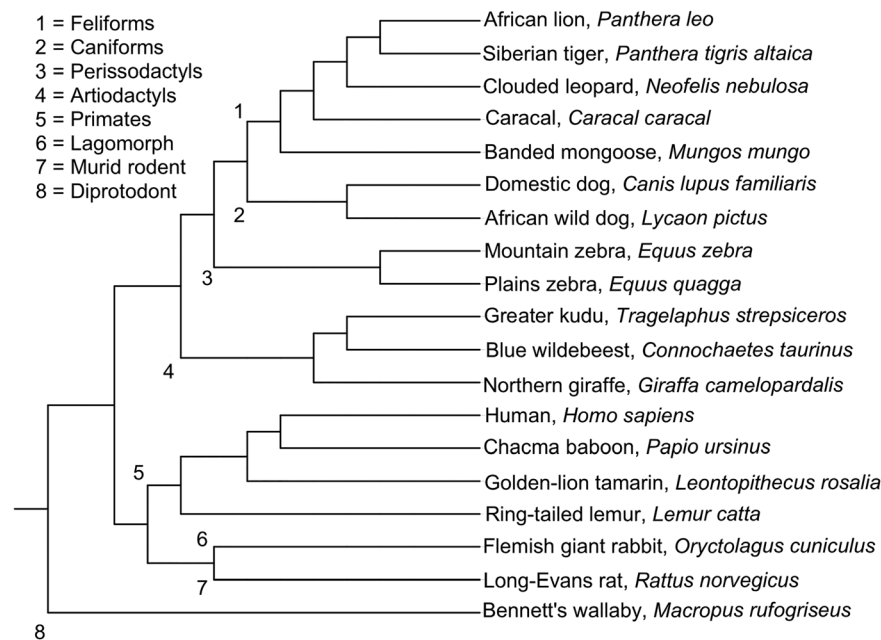


FIGURE 1 Simplified cladogram of the 19 species in the present neuromorphological analysis, organized by taxonomic grouping (as indicated in the phylogenetic tree). All species, except the wallaby, are classified as placental mammals under the magnorder Boreoeutheria (Waddell, Kishino, & Ota, 2001), which includes two superorders: Laurasiatheria and Euarchontoglires. Laurasiatheria encompasses perissodactyls, artiodactyls, feliforms, and caniforms, while Euarchontoglires includes primates, lagomorphs, and rodents (Waddell et al., 2001). Although artiodactyls (even-toed ungulates) and perissodactyls (odd-toed ungulates) are categorized within Ungulates, they are differentiated here due to molecular and phylogenetic differences (Arnason et al., 2002; Xu, Janke, & Arnason, 1996). Sequencing of mitochondrial DNA further suggests a possible sister group relationship between the carnivore and perissodactyl groups (Arnason et al., 2002; Nery et al. 2012; Waddell et al., 2001; Xu et al., 1996)

2.1.3 | Neuron selection and quantification

Traced neurons ($N = 617$) conformed to established selection criteria (Anderson et al., 2009; Jacobs et al., 2011; Jacobs, Harland, et al., 2015; Jacobs, Lee, et al., 2015; Johnson et al., 2016), with an isolated soma near the center of the 120- μm -thick section and relatively well-impregnated, unobscured, and as complete as possible (i.e., non-truncated) dendritic projections. In addition to gigantopyramidal neurons, identified by relative size and dendritic morphology (Scheibel & Scheibel, 1978a; Scheibel et al., 1974), both superficial (primarily layer III) and deep (primarily layer V) pyramidal neurons were traced for comparative purposes. Judgements of laminar determination for deep and superficial pyramidal neurons in the Golgi stain were based on the relative depth of a given neuron vis-à-vis other neurons in the same section of tissue.

Neurons were quantified under a plan achromatic 60 $\times$ oil objective along x-, y-, and z-coordinates using a Neurolucida system (MBF Bioscience, Williston, VT; RRID:nif-0000-10294), interfaced with an Olympus BH-2 microscope equipped with a Ludl XY motorized stage (Ludl Electronics, Hawthorne, NY) and a Heidenhain z-axis encoder (Schaumburg, IL). A MicroFire Digital CCD 2-Megapixel camera (Optronics, Goleta, CA) mounted on a trinocular head (model 1-L0229, Olympus, Center Valley, PA) displayed images on a 1920 $\times$ 1200 resolution Dell E248WFP 24-inch LCD monitor. Somata were first traced at their widest point in the two-dimensional plane to provide an estimate of cross-sectional area. Dendrites were then traced somatofugally in their entirety, accounting for dendritic diameter and spine quantity.

Dendritic arbors were not traced into neighboring sections; broken ends and indefinite terminations were labeled as incomplete endings. Neurons with sectioned segments were not excluded because elimination of such tracings would have biased the sample toward smaller neurons (Schadé & Caveness, 1968; Uylings, Ruiz-Marcos, & van Pelt, 1986). When present, axons were also traced, although they typically were not visible after a short distance.

Intra-rater reliability was established by having the four investigators (HKC, MEG, MET, NBSS) each trace the same soma and dendrite 10 times. The average coefficient of variation for soma size (3.3%), total dendritic length (1.2%), and dendritic spine number (4.8%) indicated little variation in tracings. Furthermore, a split plot design, ($\alpha = 0.05$) revealed no significant differences between the first and last five tracings for each investigator. Inter-rater reliability was established by comparing 10 different soma and dendrite tracings completed by each investigator with those traced by the primary investigator (BJ). Inter-class correlation averages for soma size (0.99), total dendritic length (0.99), and dendritic spine number (0.98) were not significantly different among investigators (analysis of variance, ANOVA, $\alpha = 0.05$). Additionally, the primary investigator reexamined all completed tracings under the microscope to ensure accuracy.

2.1.4 | Quantitative dendritic and spine measures

A centrifugal nomenclature was used to characterize branches extending from the soma as first-order segments, which bifurcate into second-

TABLE 1 Summary of sampled species/specimens in neuronal morphological analysis (in alphabetical order by taxonomic group)^a

Taxonomic group/species	Age (years)/ Sex	Origin ^b	Hemisphere	Brain mass (g)	Fixation method	Fixation duration ^c	Storage time ^d	Cause of death	Autolysis time
Artiodactyls									
Blue wildebeest <i>Connochaetes taurinus</i>	Adult/Male	U. of Witwatersrand South Africa	Left	306	<i>In situ</i> perfusion	2 days	2 months	Euthanized	<30 min
Blue wildebeest <i>Connochaetes taurinus</i>	Adult/Male	U. of Witwatersrand South Africa	Left	325	<i>In situ</i> perfusion	2 days	2 months	Euthanized	<30 min
Greater kudu <i>Tragelaphus strepsiceros</i>	4/Male	Blank Park Zoo	Right	180	Immersion	7 days	6 months	Euthanized	2 hours
Giraffe <i>Giraffa camelopardalis</i>	2–4/Male	U. of Witwatersrand South Africa	Left	610	<i>In situ</i> perfusion	2 days	3 months	Euthanized	<30 min
Giraffe <i>Giraffa camelopardalis</i>	2–4/Male	U. of Witwatersrand South Africa	Left	480	<i>In situ</i> perfusion	2 days	3 months	Euthanized	<30 min
Caniforms									
African wild dog <i>Lycaon pictus</i>	0.6/Male	Borås Zoo	Left	167	<i>In situ</i> perfusion	1 day	4 months	Euthanized	<30 min
African wild dog <i>Lycaon pictus</i>	0.6/Female	Borås Zoo	Left	158	<i>In situ</i> perfusion	1 day	4 months	Euthanized	<30 min
Domestic dog <i>Canis lupus familiaris</i>	1/Male	U. of Des Moines Iowa	Right	80	Immersion	3 days	5 months	Euthanized	<30 min
Diprotodont									
Bennett's wallaby <i>Macropus rufogriseus</i>	Adult/Male	Copenhagen Zoo	Left	32	<i>In situ</i> perfusion	1 day	16 months	Euthanized	<30 min
Feliforms									
Banded mongoose <i>Mungos mungo</i>	Adult/-	Copenhagen Zoo	Left	11	<i>In situ</i> perfusion	1 day	24 months	Euthanized	<30 min
Caracal <i>Caracal caracal</i>	Adult/Male	Copenhagen Zoo	Left	58	<i>In situ</i> perfusion	2 days	21 months	Euthanized	<30 min
Clouded leopard <i>Neofelis nebulosa</i>	20/Female	Smithsonian National Zoological Park	Left	82	Immersion	10 days	5 months	Euthanized	<30 min
Siberian tiger <i>Panthera tigris altaica</i>	12/Female	Copenhagen Zoo	Left	258	<i>In situ</i> perfusion	2 days	6 months	Euthanized	<30 min
African lion <i>Panthera leo</i>	Adult/ Female	Copenhagen Zoo	Left	213	<i>In situ</i> perfusion	2 days	14 months	Euthanized	<30 min
Lagomorph									
Flemish giant rabbit <i>Oryctolagus cuniculus</i>	2/Male	U. of Des Moines Iowa	Right	12	Immersion	3 days	3 months	Euthanized	<30 min

(Continues)

TABLE 1 (Continued)

Taxonomic group/species	Age (years)/ Sex	Origin ^b	Hemisphere	Brain mass (g)	Fixation method	Fixation duration ^c	Storage time ^d	Cause of death	Autolysis time
Perissodactyls									
Mountain zebra <i>Equus zebra</i>	Adult/Male	U. of Witwatersrand South Africa	Left	461	<i>In situ</i> perfusion	2 days	3 months	Euthanized	<30 min
Plains zebra <i>Equus quagga</i>	Adult/Male	Copenhagen Zoo	Left	659	<i>In situ</i> perfusion	3 days	12 months	Euthanized	<30 min
Primates									
Ring-tailed lemur <i>Lemur catta</i>	Adult/Male	Copenhagen Zoo	Left	25	<i>In situ</i> perfusion	1 day	11 months	Euthanized	<30 min
Golden lion tamarin <i>Leontopithecus rosalia</i>	Adult/Male	Copenhagen Zoo	Left	12	<i>In situ</i> perfusion	1 day	6 months	Euthanized	<30 min
Chacma baboon <i>Papio ursinus</i>	Adult/Male	U. of Witwatersrand South Africa	Left	148	<i>In situ</i> perfusion	2 days	18 months	Euthanized	<30 min
Human <i>Homo sapiens</i>	46/Male	El Paso County Coroner Colorado Springs	Left	1,388	Immersion	2 months	-	Motor Vehicle Accident	23 hr
Rodent									
Long-Evans rat <i>Rattus norvegicus</i>	0.2/Female	Colorado College Colorado Springs	Both	3	<i>In situ</i> perfusion	2 months	-	Euthanized	<30 min
Long-Evans rat <i>Rattus norvegicus</i>	0.2/Female	Colorado College Colorado Springs	Both	3	<i>In situ</i> perfusion	2 months	-	Euthanized	<30 min

^aEach animal examined is listed individually. Thus, blue wildebeest, giraffe, and Long-Evans rat species are each represented by two animals; all other species are represented by one animal.

^bBrain specimens with the "Origin" listed as the U. of Witwatersrand were collected from free-ranging animals in their natural habitat.

^cThe total time tissue was immersed in 10% neutral buffered formalin, or stored in 4% paraformaldehyde fixative after perfusion.

^dThe amount of time fixed tissue was stored in 0.1% sodium azide before sectioning.

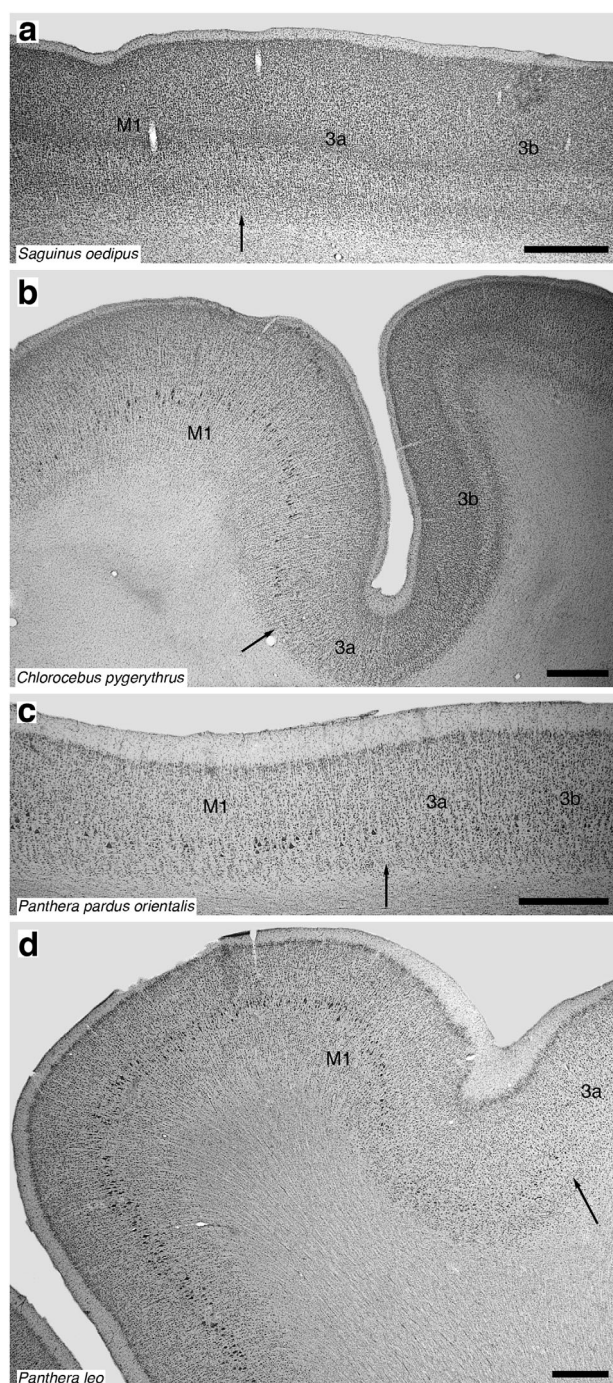


FIGURE 2 Low power photomicrographs of sagittal Nissl stained sections through the primary motor cortex (M1) illustrating its caudal border (arrow) with cortical regions posterior to M1 (3a and 3b of somatosensory cortex) in four of the species analyzed in the current study: (a) cotton-top tamarin (*Saguinus oedipus*), (b) vervet monkey (*Chlorocebus pygerythrus*), (c) Amur leopard (*Panthera pardus orientalis*), and (d) African lion (*Panthera leo*). In all images, rostral is to the left and dorsal to the top. Scale bars = 1 mm

and then third-order segments, and so on (Bok, 1959; Uylings et al., 1986). In Golgi-stained material, soma size (i.e., surface area, μm^2) and depth from the pial surface (μm) were quantified first, followed by six other measures analyzed in previous studies (Jacobs et al., 2011; Jacobs, Harland, et al., 2015; Johnson et al., 2016): dendritic volume

(Vol, μm^3 : the total volume of all dendrites); total dendritic length (TDL, μm : the summed length of all dendritic segments); mean segment length (MSL, μm : the average length of each dendritic segment); dendritic segment count (DSC: the number of dendritic segments); dendritic spine number (DSN: the total number of spines on dendritic segments); and dendritic spine density (DSD: the average number of spines per μm of dendritic length). Dendritic branching patterns were also analyzed using a Sholl (1953) analysis, which quantified dendritic intersections at 20- μm intervals radiating somatofugally. All descriptive measures are presented as mean $\pm$ standard deviation (SD) unless noted otherwise.

2.1.5 | Statistical analyses

Each neuron was examined for relative completeness immediately after it was traced. This was a subjective determination to identify neurons that were obviously incomplete vis-à-vis other neurons in the sample—for example, tracings of gigantopyramidal neurons in the clouded leopard were clearly incomplete when compared to those in the tiger. Although incomplete neurons were included in the present descriptive analysis to allow a broadly representative sample, only relatively complete neurons were used for quantitative hypothesis testing. Of the 617 neurons traced in the current study, a subset of these ($n = 523$) were categorized as being relatively complete (superficial pyramidal neurons = 220; deep pyramidal neurons = 181; gigantopyramidal neurons = 122). A full breakdown of the number of each relatively complete neuron type by taxonomic groups and species is provided in Table 2. For statistical analyses, the carnivore order was divided into the two suborders (caniforms and feliforms) because of the number of species available and because previous reports have indicated substantial somatodendritic differences between these suborders (Brodman, 1909; Johnson et al., 2016).

An initial one-way ANOVA on ranks was used as a descriptive analysis evaluating a null hypothesis test of no differences in dependent measures (i.e., Vol, TDL, MSL, DSC, DSN, DSD, and soma size) among the three neuron types. If the null hypothesis failed to be rejected, subsequent investigation would be limited to descriptive analyses; however, if the null hypothesis were rejected, additional quantitative assessment of morphological differences would be warranted. The non-normal distribution of the dependent measures necessitated the use of non-parametric Kruskal-Wallis test by ranks. Next, cluster analyses for each type of neuron were used to evaluate if taxonomic groups were differentiable by the dependent measures. Analyses for superficial, deep, and gigantopyramidal neurons proceeded with one cluster for each taxonomic group. The minimum for cluster solution validity was set at an overall 75% correct classification of taxonomic group. Parameterization of the cluster analysis included 100 iterations using squared Euclidian distances and standardized variable values. Concomitant ANOVAs were performed to evaluate if the composite of neuronal measures discriminated among taxonomic groups in the cluster analyses.

2.2 | Stereological analysis

2.2.1 | Specimens

The brains from 11 adult carnivore and nine primate species—one specimen for each species—were acquired as they became available from

TABLE 2 Numerical breakdown of relatively complete neurons by taxonomic group and species used in statistical analysis of neuronal morphology

Taxonomic Group	Species	Number of neurons	Superficial by species	Superficial by group	Deep by Species	Deep by group	Gigantopyramidal by species	Gigantopyramidal by group
Artiodactyls	Wildebeest	53	19		13		21	
	Giraffe	26	11	43	10	34	5	29
	Kudu	27	13		11		3	
Caniforms	African wild dog	77	40	49	29	35	8	14
	Domestic dog	21	9		6		6	
Diprotodont	Wallaby	18	9	9	9	9	0	0
Feliforms	Caracal	24	10		9		5	
	Leopard	6	5		1		0	
	Lion	29	10	41	10	36	9	17
	Mongoose	23	10		13		0	
	Tiger	12	6		3		3	
Lagomorph	Rabbit	27	14	14	13	13	0	0
Perissodactyls	Mountain zebra	21	5	15	6	17	10	17
	Plains zebra	28	10		11		7	
Primates	Baboon	29	10		10		9	
	Human	37	10		10		17	
	Lemur	10	9	39	1	31	0	45
	Tamarin	39	10		10		19	
Rodent	Rat	16	10	10	6	6	0	0
Total		523	220		181		122	

the Copenhagen Zoo, Denmark (Table 3). Animals were obtained after being euthanized with sodium pentobarbital (i.v. for larger animals; i.p. for smaller animals) in line with veterinary and management protocols of the zoo. The heads of the larger animals were removed from the body and perfused through the carotid arteries with a rinse of 0.9% saline (1 l/2 kg tissue mass), followed by fixation of 4% paraformaldehyde in 0.1 M phosphate buffer (PB; 1 l/1 kg tissue mass; Manger et al., 2009). The bodies of smaller animals were perfused through the heart using the same solutions and amounts of solution per kg of body mass. The brains were removed from the skull and post-fixed (in 4% paraformaldehyde in 0.1 M PB) for 24–72 hr, depending on brain mass, at 4°C. The brains were then transferred to a solution of 30% sucrose in 0.1 M PB at 4°C until they had equilibrated and were then immersed in an antifreeze solution containing 30% glycerol, 30% ethylene glycol, 30% distilled water, and 10% 0.244 M PB. The brains were allowed to equilibrate in the antifreeze solution at 4°C and then moved to a –20°C freezer for storage.

2.2.2 | Tissue selection and sectioning

As with the Golgi neuromorphological analysis, blocks of tissue from the mid-dorsolateral region of M1 were dissected from the brains of each species (Figure 2). Stereological measurements were taken from

this region, which included high numbers of gigantopyramidal neurons. Tissue blocks were frozen in dry ice, attached to an aluminum stage and sectioned at 50 µm thickness in a coronal plane on a sliding microtome. A minimum of six sections per block were mounted on 0.5% gelatin-coated glass slides and then cleared in a solution of 1:1 chloroform and 100% alcohol overnight, and then stained with 1% cresyl violet.

2.2.3 | Quantification protocol and statistical analysis

For the quantification of cortical pyramidal and gigantopyramidal neurons, an unbiased design was employed based on a systematic random sampling stereological protocol with strict inclusion criteria (e.g., only whole cell bodies). We used a MicroBrightfield (MBF Bioscience, Williston, VT; RRID:nif-0000-10294) system with a three-plane motorized stage, Zeiss.Z2 Vario Axioimager (63× oil objective) and StereoInvestigator software (version 11.08.1; 64-bit). The nucleator probe of StereoInvestigator was used to estimate the length, area, and volume of somata. The number of neurons measured is provided in Table 3. Separate pilot studies for length, area, and volume for each neuron type from Nissl-stained sections were conducted to optimize sampling parameters, such as the counting frame and sampling grid sizes, and to achieve a coefficient of error (CE) below 0.1 (Dell, Patzke, Spocter,

TABLE 3 Estimated average lengths, areas, and volumes of layer V pyramidal and gigantopyramidal neuron somata in carnivore and primate species (in alphabetical order by species)^a

Species	Body mass (kg)	Brain mass (g)	Pyramidal neuron lengths (μm)			Gigantopyramidal neuron lengths (μm)			Pyramidal neuron areas (μm ²)			Gigantopyramidal neuron areas (μm ²)			Pyramidal neuron volumes (μm ³)			Gigantopyramidal neuron volumes (μm ³)		
			n	Mean	Range	n	Mean	Range	n	Mean	Range	n	Mean	Range	n	Mean	Range	n	Mean	Range
Carnivores																				
African lion <i>Panthera leo</i>	155.8	223.5	137	8.9 ± 1.5	5.7–14.8	15	20.5 ± 4.6	15.0–29.8	134	268.9 ± 86.1	113.3–596.0	16	1,416.2 ± 644.4	728.6–2,858.5	132	3,793.6 ± 1,872.5	1,014.5–13,002.4	16	46,173.6 ± 30,550.2	15,350.5–119,782.3
African wild dog <i>Lycan pictus</i>	26.3	141.5	178	9.1 ± 1.0	6.3–10.8	83	12.9 ± 2.0	10.8–21.7	167	286.4 ± 51.8	20.11–389.4	76	586.4 ± 206.7	391.0–1,622.1	207	4,391.3 ± 1,700.0	2,025.6–8,996.72	41	16,074.3 ± 8,689.9	9,109.7–55,734.1
Amur leopard <i>Panthera pardus orientalis</i>	52.4	125.5	292	8.3 ± 1.1	6.2–11.7	29	19.2 ± 6.1	12.3–29.4	295	225.9 ± 64.1	122.4–458.2	29	1,316 ± 818.2	479.5–2,765.8	298	2,775.7 ± 1,291.9	1,051.0–8,841.8	27	45,239.7 ± 38,077.4	9,888.8–117,188.8
Asian small-clawed otter <i>Amblonyx cinereus</i>	3.5	38.1	216	8.6 ± 1.6	6.2–12.8	31	16.0 ± 2.5	13.1–23.0	217	252.1 ± 102.2	124.5–552.5	29	872.8 ± 274.0	566.7–1,667.0	218	3,266.5 ± 2,056.3	1,006.1–9,897.8	30	20,560.2 ± 10,114.9	10,167.7–51,610.2
Banded mongoose <i>Mungos mungo</i>	1.3	10.5	503	7.5 ± 0.8	6.0–9.0	60	9.8 ± 0.8	9.0–12.0	509	187.5 ± 38.1	120.0–278.3	46	330.3 ± 51.4	281.3–495.8	509	2,099.2 ± 691.7	1,003.2–3,590.5	31	5,297.7 ± 1,167.3	40,161.1–8,954.5
Caracal <i>Caracal caracal</i>	11.6	55.3	282	8.8 ± 1.2	7.6–12.3	41	15.2 ± 2.5	12.7–27.7	285	258.3 ± 74.8	186.0–520.0	41	775.0 ± 317.0	520.9–2,509.1	284	3,415.1 ± 1,663.6	2,001.3–9,691.3	40	18,421.7 ± 14,309.5	10,129.5–100,697.6
European polecat <i>Mustela putorius</i>	1.1	8.3	177	8.2 ± 1.5	6.0–11.0	44	12.4 ± 1.0	11.1–14.8	183	241.1 ± 89.3	119.3–433.3	32	533.6 ± 68.6	443.5–725.2	190	3,214.3 ± 1,819.4	1,004.3–7,649.7	27	10,231.0 ± 1,777.3	8,187.3–15,925.5
Harp seal <i>Pagophilus groenlandicus</i>	132.3	276.0	240	9.1 ± 1.2	6.3–11.0	49	11.7 ± 0.7	11.0–13.9	237	271.7 ± 65.2	127.9–387.9	52	438.5 ± 54.3	390.5–614.6	237	3,585.2 ± 1,233.7	1,104.6–5,983.4	52	7,296.3 ± 1,491.9	6,012.3–11,984.2
Northern fur seal <i>Callorhinus ursinus</i>	135.9	328.6	433	9.6 ± 1.6	6.1–12.5	43	13.4 ± 1.0	12.5–17.5	447	319.1 ± 105.6	122.3–549.2	27	642.8 ± 106.1	556.4–1,053.3	443	4,667.7 ± 2,249.8	1,013.7–9,971.2	34	12,955.6 ± 3,638.7	10,033.7–28,902.2
Raccoon <i>Procyon lotor</i>	6.4	40.0	541	8.8 ± 1.2	6.3–13.0	39	15.4 ± 2.3	13.0–21.5	551	254.1 ± 74.8	120.7–548.4	37	794.6 ± 253.8	562.4–1,488.5	550	3,232.9 ± 1,440.6	1,000.8–9,666.8	40	17,746.9 ± 9,299.3	10,075.0–44,546.1
Siberian tiger <i>Panthera tigris altaica</i>	161.0	279.3	126	8.6 ± 1.2	6.4–11.8	12	17.0 ± 4.6	12.1–27.8	128	245.9 ± 72.7	126.3–452.0	12	1,040.0 ± 569.8	460.9–2,517.8	131	3,158.4 ± 14,71.9	1,045.7–7,560.3	11	32,289.6 ± 25,103.6	9,909.7–99,838.6
Primates																				
Black-capped squirrel monkey <i>Saimiri boliviensis</i>	0.9	25.5	146	6.5 ± 0.5	5.8–7.5	24	8.2 ± 0.8	7.7–11.1	126	140.7 ± 20.2	114.2–189.5	22	224.4 ± 46.4	191.8–392.0	109	1,360.5 ± 266.4	1,001.7–1,949.0	24	2,646.0 ± 935.7	2,004.6–6,030.8
Chacma baboon <i>Papio ursinus</i>	31.0	214.4	27	6.5 ± 0.2	6.0–6.9	27	8.0 ± 0.8	7.2–9.8	35	143.8 ± 16.3	121.0–178.3	19	223.7 ± 35.8	182.0–305.8	38	1,364.1 ± 252.0	1,016.7–1,909.4	18	2,662.7 ± 593.6	2,014.3–4,069.2
Cotton-top tamarin <i>Saguinus oedipus</i>	0.4	8.9	137	6.5 ± 0.5	5.8–7.5	35	8.9 ± 1.2	7.6–12.0	108	145.5 ± 18.3	118.3–187.9	37	260.9 ± 73.5	191.9–473.5	111	1,368.7 ± 239.2	1,000.1–1,955.0	41	3,349.5 ± 1,506.8	2,008.8–8,339.8
Golden lion tamarin <i>Leontopithecus rosalia</i>	0.6	12.8	134	5.7 ± 0.5	4.8–6.7	27	7.5 ± 0.7	6.7–9.5	130	107.7 ± 19.8	77.0–148.1	32	183.1 ± 38.5	149.5–299.5	140	909.3 ± 272.5	511.5–1,498.9	29	2,171.0 ± 713.3	1,530.6–4,147.9

(Continues)

TABLE 3 (Continued)

Species	Body mass (kg)	Brain mass (g)	Pyramidal neuron lengths (μm)			Gigantopyramidal neuron lengths (μm)			Pyramidal neuron areas (μm ²)			Gigantopyramidal neuron areas (μm ²)			Pyramidal neuron volumes (μm ³)			Gigantopyramidal neuron volumes (μm ³)		
			n	Mean	Range	n	Mean	Range	n	Mean	Range	n	Mean	Range	n	Mean	Range	n	Mean	Range
Hamadryas baboon	26.0	159.1	104	6.5 ± 0.3	6.0–7.0	64	7.9 ± 1.3	7.0–15.7	130	143.1 ± 16.0	118.3–175.0	41	235.0 ± 101.1	176.1–784.4	143	1,385.2 ± 267.5	1,002.6–1,980.9	35	3,327.7 ± 2,644.4	2,001.8–16,828.2
<i>Papio hamadryas</i>																				
Lar gibbon	5.5	134.8	97	7.8 ± 0.1	6.1–8.6	96	9.3 ± 0.04	8.6–9.9	123	205.8 ± 30.6	122.7–247.9	46	273.8 ± 13.6	251.8–295.1	159	2,558.9 ± 637.9	1,061.1–3,640.8	71	5,603.9 ± 3,121.4	3,657.9–12,964.4
<i>Hylobates lar</i>																				
Pygmy marmoset	0.1	4.1	108	6.6 ± 0.3	6.2–7.4	42	8.3 ± 0.6	7.5–9.6	127	141.3 ± 16.4	120.0–183.7	40	229.4 ± 35.0	187.4–295.0	134	1,304.5 ± 229.8	1,002.2–1,943.7	42	2,757.2 ± 666.1	2,007.3–4,313.6
<i>Cebuella pygmaea</i>																				
Red-tailed monkey	4.2	58.4	193	6.7 ± 0.3	6.2–7.4	111	8.1 ± 0.6	7.4–10.5	241	150.2 ± 20.2	120.7–189.2	76	224.5 ± 27.7	190.1–347.6	239	1,421.9 ± 287.1	1,000.6–1,992.0	83	2,567.1 ± 490.8	2,002.7–4,901.5
<i>Cercopithecus ascanius</i>																				
Vervet monkey	5.6	71.8	57	7.2 ± 0.6	6.2–8.4	26	10.3 ± 1.9	8.4–14.9	62	167.1 ± 30.4	122.0–228.9	27	352.6 ± 139.9	237.6–715.6	71	1,698.7 ± 536.0	1,005.8–2,929.2	26	5,636.5 ± 3,458.1	3,033.8–15,219.6
<i>Chlorocebus pygerythrus</i>																				

^aEach species is represented by one animal. The standard deviation and the range observed as well as the number of cells measured are provided. Estimates of area were calculated using the “nucleator” probe of the StereoInvestigator software (see text for details).

Siegel, & Manger, 2016; Gundersen, 1988; West, Slomianka, & Gundersen, 1991). In addition, we measured the tissue section thickness at every 10th sampling site; the vertical guard zone was determined according to tissue thickness to avoid errors or biases due to sectioning artifacts (Dell et al., 2016; West et al., 1991). All specimens were processed under the same conditions at the same time to minimize variation in shrinkage of tissue samples and to reduce unfavorable stereological bias. With regard to the vertical guard zones, uniform vertical distances were applied to all tissue samples (Table 4). Table 4 provides a detailed summary of the stereological parameters used in each species examined. During quantification, the observer (AB) blindly measured both cortical pyramidal and gigantopyramidal neurons. Once the stereological measurements were completed, the data were then distributed in a bi-modal manner to distinguish cortical pyramidal neurons from gigantopyramidal neurons. The stereologically observed ranges of each neuronal type for each recorded parameter are reported in Table 3.

To compare the bivariate scaling relationships of body mass and brain mass against the size dimensions for pyramidal and gigantopyramidal neuronal somata, we performed linear regression analyses (Reduced Major Axis, RMA) using the statistical package PAST (Version 3.14; PAST © Hammer 1999–2016; Hammer, Harper, & Ryan, 2001). Because of the markedly large soma size of the gigantopyramidal neurons in genus *Panthera* (i.e., African lion, Amur leopard, Siberian tiger), the data were divided into three groups for subsequent analyses: primates, non-*Panthera* carnivores (non-P carnivores), and *Panthera*. All statistical analyses were conducted on logarithmic (base 10) transformed data with linear models fit to the non-P carnivores and primate (sub)orders. To test for significant differences in size parameters between groups (e.g., non-P carnivores vs. primates; *Panthera* vs. non-P carnivores), we used an ANOVA followed by post-hoc multiple comparisons (Games-Howell) for cases where significant overall effects were observed.

3 | RESULTS

3.1 | Neuronal morphology

3.1.1 | Overview

As is typical in neocortical Golgi preparations (Anderson et al., 2009), superficial layers tended to exhibit clearer backgrounds and more complete, unobstructed neurons than did deeper cortical layers. This resulted in tracings of more superficial pyramidal neurons ($n = 233$) than of deep pyramidal neurons ($n = 203$) or gigantopyramidal neurons ($n = 181$). Descriptive findings for all species are provided in Table 5, which indicates that most somatodendritic measures, especially Vol and soma size, tended to increase from superficial to deep to gigantopyramidal neurons. Across species, there was a 463-fold increase in brain mass from the smallest (rat, 3 g) to the largest brain (human, 1,388 g) in the analysis (Table 1). Variation in somatodendritic measures, however, was much less—generally, between 2-fold and 9-fold for each dependent measure across species (Table 5). To examine this relationship further, correlations were run between the mass of each

TABLE 4 Stereological parameters used for estimating neuron somata dimensions in carnivore and primate species (in alphabetical order by species)

Species	Counting frame size (μm)	Grid frame size (μm)	Disector height (μm)	Section thickness (μm)	Average mounted thickness (μm)	Guard zones (μm)	Section interval ^a	Average number of sections	Average number of sampling sites	Average CE (Gundersen $m = 1$)
Carnivores										
African lion <i>Panthera leo</i>	125 × 125	350 × 350	10	50	15.2	2	1	11	175	0.08
African wild dog <i>Lycaon pictus</i>	250 × 250	600 × 600	22	50	26.5	2	20	4	94	0.07
Amur leopard <i>Panthera pardus orientalis</i>	100 × 100	300 × 300	10	50	20.0	2	1	12	127	0.05
Asian small-clawed otter <i>Amblonyx cinereus</i>	100 × 100	275 × 275	10	50	20.0	2	1	11	153	0.06
Banded mongoose <i>Mungos mungo</i>	75 × 75	225 × 225	10	50	19.8	2	1	12	161	0.04
Caracal <i>Caracal caracal</i>	100 × 100	300 × 300	10	50	19.7	2	1	12	157	0.04
European polecat <i>Mustela putorius</i>	100 × 100	225 × 225	10	50	19.9	2	1	11	198	0.06
Harp seal <i>Pagophilus groenlandicus</i>	100 × 100	350 × 350	10	50	20.0	2	1	12	109	0.06
Northern fur seal <i>Callorhinus ursinus</i>	100 × 100	275 × 275	10	50	16.1	2	1	12	159	0.05
Raccoon <i>Procyon lotor</i>	100 × 100	275 × 275	10	50	20.0	2	1	11	199	0.04
Siberian tiger <i>Panthera tigris altaica</i>	125 × 125	375 × 375	10	50	20.6	2	1	12	147	0.07
Primates										
Black-capped squirrel monkey <i>Saimiri boliviensis</i>	75 × 75	225 × 225	15	50	20.6	2	1	12	113	0.05
Chacma baboon <i>Papio ursinus</i>	100 × 100	300 × 300	10	50	18.3	2	1	12	177	0.06
Cotton top tamarin <i>Saguinus oedipus</i>	100 × 100	200 × 200	10	50	19.6	2	1	12	229	0.05
Golden lion tamarin <i>Leontopithecus rosalia</i>	75 × 75	250 × 250	15	50	16.9	2	1	12	79	0.07
Hamadryas baboon <i>Papio hamadryas</i>	100 × 100	250 × 250	10	50	19.5	2	1	12	218	0.05

(Continues)

TABLE 4 (Continued)

Species	Counting frame size (μm)	Grid frame size (μm)	Disector height (μm)	Section thickness (μm)	Average mounted thickness (μm)	Guard zones (μm)	Section interval ^a	Average number of sections	Average number of sampling sites	Average CE (Gundersen $m = 1$)
Lar gibbon <i>Hylabates lar</i>	250 × 250	600 × 600	19	50	23.5	2	20	3	121	0.07
Pygmy marmoset <i>Cebuella pygmaea</i>	75 × 75	225 × 225	15	50	20.3	2	1	12	138	0.05
Red-tailed monkey <i>Cercopithecus ascanius</i>	100 × 100	250 × 250	10	50	19.4	2	1	12	189	0.04
Vervet monkey <i>Chlorocebus pygerythrus</i>	75 × 75	325 × 325	15	50	19.7	2	1	12	124	0.09

^aThe African wild dog and Lar gibbon samples were obtained from ongoing studies involving detailed neuroanatomical descriptions, where each brain was coronally sectioned at 50 μm in a 1:20 series of sections. Therefore, the serial section interval of 20 was different in these two species compared to all other species in the present sample, where blocks of tissue containing the primary motor cortex were sectioned at 50 μm and stained only for Nissl substance.

brain and the average Vol, TDL, and soma size values of the three neuron types. In general, somatodendritic measures increased with brain mass only in superficial pyramidal neurons, Vol: $r_{(23)} = .53$, $p = .01$; TDL: $r_{(23)} = .44$, $p = .035$; soma size: $r_{(23)} = .53$, $p = .01$. However, removing the human brain from the analysis as a potential outlier revealed a slightly stronger positive relationship between certain somatodendritic measures and brain mass, Vol: $r_{(22)} = .69$, $p = .001$; soma size: $r_{(22)} = .48$, $p = .023$, and between somatodendritic measures and deep pyramidal neurons, Vol: $r_{(21)} = .58$, $p = .006$; TDL: $r_{(21)} = .515$, $p = .017$, but not gigantopyramidal neurons.

Although gigantopyramidal dendrites were variably truncated within the 120-μm section thickness, the quality of the Golgi impregnation across all species remained relatively high, as illustrated in photomicrographs of gigantopyramidal neurons in feliforms (Figure 3), caniforms (Figure 4), perissodactyls (Figure 5), artiodactyls (Figure 6), and primates (Figure 7). Sample tracings of the three neuron types are also provided for each species: caracal (Figure 8), Siberian tiger and clouded leopard (Figure 9), African lion (Figure 10), mongoose (Figure 11), African wild dog (Figure 12), domestic dog (Figure 13), mountain zebra (Figure 14), plains zebra (Figure 15), giraffe (Figure 16), blue wildebeest and greater kudu (Figure 17), chacma baboon (Figure 18), human (Figure 19), golden lion tamarin and ring-tailed lemur (Figure 20), giant Flemish rabbit and Long-Evans rat (Figure 21), and Bennett's wallaby (Figure 22). Mean values for all somatodendritic quantifications are presented by taxonomic groups in Figure 23, and a Sholl analysis for the three neuron types across taxonomic groups is provided in Figure 24. Below, we examine the morphological characteristics of each neuron type in more detail.

3.1.2 | Superficial pyramidal neurons

Located at an average soma depth of 739 ± 15 μm, superficial pyramidal neurons expressed considerable morphological uniformity across species. Somata were characterized by a triangular or rounded shape, with diffuse, rather symmetrical basilar dendritic skirts containing an average of 4.96 ± 1.44 primary basilar dendrites. There was some variability in the apical dendrite, as bifurcations into daughter branches appeared to be more common in artiodactyls (giraffe: Figure 16e, f; wildebeest: Figure 17d–f; kudu: Figure 17m–o) and perissodactyls (mountain zebra: Figure 14d–g; plains zebra: Figure 15e, f) than in primates (baboon: Figure 18e–g; human: Figure 19e, f; golden lion tamarin: Figure 20d–f), or rabbit and rat (Figure 21a–d, i–k), where there was usually a single apical shaft with smaller oblique extensions. Quantitatively, superficial pyramidal somata were typically smaller than those of deep and gigantopyramidal neurons in each species (Table 5). Overall dendritic measures (e.g., Vol and TDL) were lower in superficial pyramidal neurons compared to deep and gigantopyramidal neurons. DSC measures were slightly higher, as were the two spine measures (e.g., DSN and DSD), in superficial pyramidal neurons than in the other two neuron types. The dependent measure with the greatest variation across species for superficial pyramidal neurons was Vol (7.62-fold; Table 5). TDL measurements (Figure 23d) illustrated less extensive somatodendritic measures in the rabbit and rat, and most extensive in feliforms and primates. Finally, in the Sholl analysis, superficial pyramidal neurons exhibited a similar pattern across all

TABLE 5 Summary statistics for each neuron type across sampled species in morphological analysis (in alphabetical order by taxonomic group)

Neuron type	n ^a	Vol ^b	TDL ^c	MSL ^d	DSC ^e	DSN ^f	DSD ^g	SoSize ^h	SoDepth ⁱ
Artiodactyls									
Blue wildebeest									
Superficial	19	13,411 ± 1,527	5,640 ± 414	77 ± 3	75 ± 6	2,224 ± 220	0.30 ± 0.02	425 ± 30	759 ± 68
Deep	13	22,158 ± 2,526	6,086 ± 306	86 ± 4	73 ± 5	2,236 ± 232	0.30 ± 0.03	479 ± 38	1,387 ± 105
Gigantopyramidal	26	51,400 ± 3,386	5,666 ± 266	101 ± 4	57 ± 3	3,244 ± 219	0.50 ± 0.02	882 ± 45	1,361 ± 47
Giraffe									
Superficial	11	16,059 ± 2,157	5,177 ± 319	81 ± 3	64 ± 3	4,955 ± 383	0.84 ± 0.06	301 ± 27	658 ± 31
Deep	10	16,222 ± 1,927	5,980 ± 659	85 ± 6	70 ± 5	4,370 ± 579	0.58 ± 0.03	408 ± 31	1,218 ± 37
Gigantopyramidal	5	73,427 ± 15,718	8,397 ± 903	100 ± 4	84 ± 8	7,794 ± 902	0.75 ± 0.04	1,143 ± 112	1,513 ± 92
Kudu									
Superficial	16	7,445 ± 633	3,655 ± 269	71 ± 3	51 ± 3	2,068 ± 238	0.40 ± 0.04	323 ± 13	948 ± 18
Deep	13	10,316 ± 1,309	3,771 ± 244	72 ± 5	54 ± 4	1,726 ± 199	0.30 ± 0.03	357 ± 30	1,468 ± 91
Gigantopyramidal	7	25,210 ± 2,779	5,191 ± 592	87 ± 4	61 ± 9	2,135 ± 173	0.30 ± 0.04	745 ± 71	1,724 ± 102
Caniforms									
African wild dog									
Superficial	41	10,853 ± 677	4,681 ± 222	79 ± 2	60 ± 3	2,343 ± 147	0.40 ± 0.02	376 ± 18	759 ± 26
Deep	31	11,050 ± 639	4,707 ± 259	83 ± 3	57 ± 3	1,892 ± 193	0.30 ± 0.02	428 ± 18	1,385 ± 62
Gigantopyramidal	9	27,317 ± 3,821	5,794 ± 499	79 ± 4	72 ± 5	3,038 ± 399	0.42 ± 0.04	2,662 ± 261	1,554 ± 45
Domestic dog									
Superficial	12	9,446 ± 1,147	3,112 ± 165	58 ± 3	55 ± 4	929 ± 95	0.30 ± 0.02	430 ± 44	718 ± 41
Deep	8	12,572 ± 1,700	3,429 ± 366	68 ± 5	51 ± 5	929 ± 122	0.20 ± 0.03	629 ± 67	1,230 ± 75
Gigantopyramidal	17	37,564 ± 3,807	3,524 ± 238	73 ± 4	49 ± 3	1,225 ± 149	0.30 ± 0.03	1,876 ± 187	1,402 ± 66
Diprotodont									
Bennett's wallaby									
Superficial	10	4,951 ± 636	4,739 ± 381	90 ± 5	53 ± 4	2,697 ± 298	0.48 ± 0.04	258 ± 13	540 ± 37
Deep	13	14,151 ± 2,023	4,705 ± 619	85 ± 6	52 ± 4	2,753 ± 467	0.38 ± 0.03	437 ± 31	1,052 ± 33
Feliforms									
Banded mongoose									
Superficial	10	6,400 ± 1,010	5,092 ± 539	72 ± 5	71 ± 6	3,011 ± 442	0.48 ± 0.04	279 ± 21	622 ± 29
Deep	14	11,551 ± 1,404	4,931 ± 327	81 ± 5	61 ± 3	3,120 ± 251	0.51 ± 0.03	484 ± 74	992 ± 34
Caracal									
Superficial	10	7,962 ± 1,117	6,008 ± 464	82 ± 4	75 ± 6	3,395 ± 470	0.46 ± 0.05	370 ± 29	670 ± 54
Deep	9	15,765 ± 3,052	5,353 ± 329	88 ± 5	61 ± 3	3,661 ± 306	0.53 ± 0.02	448 ± 45	1,267 ± 64
Gigantopyramidal	10	67,799 ± 15,311	7,222 ± 712	88 ± 4	81 ± 6	1,187 ± 214	0.15 ± 0.05	2,000 ± 182	1,199 ± 47
Clouded leopard									
Superficial	5	8,452 ± 1,563	3,521 ± 255	62 ± 7	59 ± 6	1,353 ± 37	0.35 ± 0.02	306 ± 27	571 ± 50
Deep	1	4,203	4,436	78	57	1,170	0.22	294	1,435
Gigantopyramidal	6	61,253 ± 11,929	1,179 ± 205	63 ± 7	18 ± 2	240 ± 38	0.20 ± 0.03	3,720 ± 352	1,363 ± 53
Siberian tiger									
Superficial	6	20,182 ± 5,677	6,215 ± 247	83 ± 8	77 ± 5	2,257 ± 308	0.28 ± 0.04	439 ± 85	752 ± 71
Deep	3	16,066 ± 1,812	5,589 ± 352	97 ± 14	60 ± 10	1,496 ± 98	0.19 ± 0.02	350 ± 62	1,486 ± 244
Gigantopyramidal	8	94,701 ± 27,983	5,513 ± 1,003	128 ± 7	42 ± 6	1,005 ± 157	0.16 ± 0.02	2,844 ± 366	1,546 ± 29
African lion									
Superficial	10	9,762 ± 1,747	4,998 ± 259	77 ± 3	65 ± 3	1,415 ± 159	0.23 ± 0.03	358 ± 40	799 ± 45
Deep	10	17,673 ± 4,006	5,800 ± 555	83 ± 4	69 ± 5	1,908 ± 306	0.24 ± 0.02	525 ± 87	1,376 ± 73
Gigantopyramidal	10	88,381 ± 8,281	8,600 ± 814	120 ± 8	72 ± 6	1,099 ± 154	0.10 ± 0.01	2,824 ± 284	1,558 ± 40
Lagomorph									
Flemish giant rabbit									
Superficial	19	4,261 ± 299	2,861 ± 207	67 ± 2	43 ± 3	1,365 ± 113	0.40 ± 0.02	248 ± 12	762 ± 27
Deep	18	6,353 ± 797	3,522 ± 313	66 ± 3	52 ± 4	1,115 ± 125	0.20 ± 0.02	350 ± 19	1,322 ± 34
Perissodactyls									
Mountain zebra									
Superficial	5	12,487 ± 1,144	5,352 ± 529	92 ± 4	58 ± 3	2,221 ± 701	0.30 ± 0.08	423 ± 31	796 ± 57
Deep	6	17,925 ± 3,584	5,433 ± 524	93 ± 5	60 ± 7	2,109 ± 375	0.30 ± 0.03	464 ± 46	1,427 ± 131
Gigantopyramidal	21	60,377 ± 3,604	4,871 ± 340	89 ± 2	55 ± 4	2,280 ± 202	0.40 ± 0.03	1,260 ± 70	1,434 ± 60
Plains zebra									
Superficial	10	13,088 ± 1,963	5,105 ± 469	79 ± 1	65 ± 6	2,896 ± 497	0.45 ± 0.04	377 ± 42	818 ± 50
Deep	13	14,339 ± 1,613	5,015 ± 407	87 ± 4	58 ± 5	2,637 ± 307	0.41 ± 0.02	455 ± 28	1,168 ± 59
Gigantopyramidal	15	51,879 ± 5,012	3,970 ± 395	104 ± 4	39 ± 4	1,976 ± 271	0.37 ± 0.03	870 ± 79	1,188 ± 21

(Continues)

TABLE 5 (Continued)

Neuron type	n ^a	Vol ^b	TDL ^c	MSL ^d	DSC ^e	DSN ^f	DSD ^g	SoSize ^h	SoDepth ⁱ
Primates									
Ring-tailed lemur									
Superficial	9	5,729 ± 672	4,571 ± 365	65 ± 3	70 ± 5	2,652 ± 329	0.50 ± 0.03	225 ± 14	526 ± 33
Deep	1	2,611	3,118	82	38	1,235	0.28	172	877
Gigantopyramidal	2	22,159 ± 5,598	4,787 ± 691	69 ± 0.04	69 ± 10	2,308 ± 200	0.46 ± 0.03	1,007 ± 37	1,188 ± 39
Golden lion tamarin									
Superficial	10	2,650 ± 512	4,476 ± 330	64 ± 2	71 ± 6	1,986 ± 160	0.42 ± 0.03	185 ± 21	715 ± 52
Deep	10	5,111 ± 856	5,029 ± 447	70 ± 4	72 ± 6	2,211 ± 257	0.38 ± 0.03	268 ± 34	1,255 ± 76
Gigantopyramidal	19	26,474 ± 2,508	6,286 ± 399	89 ± 3	73 ± 6	2,008 ± 123	0.28 ± 0.02	954 ± 60	1,230 ± 38
Chacma baboon									
Superficial	10	8,294 ± 1,142	6,511 ± 494	68 ± 3	95 ± 5	4,468 ± 546	0.70 ± 0.03	282 ± 21	615 ± 54
Deep	10	10,491 ± 2,355	5,968 ± 615	77 ± 4	77 ± 6	3,347 ± 387	0.50 ± 0.03	389 ± 59	1,505 ± 114
Gigantopyramidal	9	45,638 ± 10,333	9,524 ± 621	90 ± 4	107 ± 7	4,002 ± 837	0.40 ± 0.05	1,018 ± 167	1,763 ± 157
Human									
Superficial	10	12,717 ± 1,458	6,121 ± 487	66 ± 3	92 ± 4	2,213 ± 227	0.32 ± 0.02	452 ± 46	1,255 ± 97
Deep	10	11,300 ± 1,949	5,329 ± 516	80 ± 3	68 ± 7	2,108 ± 313	0.33 ± 0.02	454 ± 60	2,146 ± 96
Gigantopyramidal	17	29,412 ± 2,804	7,706 ± 503	91 ± 4	87 ± 7	1,736 ± 275	0.20 ± 0.03	969 ± 52	1,876 ± 68
Rodent									
Rat									
Superficial	10	4,126 ± 346	3,785 ± 257	61 ± 2	62 ± 4	2,252 ± 184	0.50 ± 0.02	173 ± 11	468 ± 22
Deep	10	10,015 ± 1,153	4,605 ± 262	71 ± 3	65 ± 4	2,174 ± 198	0.41 ± 0.03	362 ± 22	1,208 ± 47

^aNumber of cells traced.^bVolume in μm^3 .^cLength in μm .^dAverage length of dendritic segments in μm .^eNumber of segments per neuron.^fNumber of spines per neuron.^gNumber of spines per μm of dendritic length.^hSoma size in μm^2 .ⁱSoma depth in μm from the pial surface.

taxonomic groups, with a peak in intersections ~ 100 – $140 \mu\text{m}$ from the soma; however, this peak was higher in primates and feliforms than in other species (Figure 24).

3.1.3 | Deep pyramidal neurons

Deep pyramidal neurons resided mostly in layer V (with a few possibly in layer VI), with an average soma depth of $1,331 \pm 24 \mu\text{m}$. Deep pyramidal neurons generally resembled their superficial counterparts in having conical shaped somata. Basilar dendritic geometry differed, however, insofar as it appeared to be more variable and less symmetrical than in superficial pyramidal neurons. The average number of primary basilar dendrites (5.21 ± 1.78) for deep pyramidal neurons was also somewhat greater than in superficial pyramidal neurons. As with superficial pyramidal neurons, apical dendrites generally exhibited fewer bifurcations in primates, the rabbit, and the rat. However, a candelabra-like apical dendritic morphology appeared in the mongoose (Figure 11d, e) and the wallaby (Figure 22d–f), where deep pyramidal neurons exhibited apical dendrites that bifurcated soon after leaving the soma, with daughter branches ascending to the pial surface nearly parallel to each other. Quantitatively, all dependent measures for deep pyramidal neurons (except DSC, which was lower) were between those of superficial and gigantopyramidal neurons. The dependent measure with the greatest variation across species for deep pyramidal neurons was Vol (8.49-fold; Table 5). Across species, the somatodendritic measures of deep pyramidal neurons tended to be less extensive in the

rabbit, rat, and caniforms, and relatively more extensive in artiodactyls, perissodactyls, primates, feliforms, and primates, as illustrated by TDL measurements (Figure 23e). The Sholl analysis revealed that deep pyramidal neurons exhibited a similar pattern to that observed in superficial pyramidal neurons, albeit with greater lateral dendritic extension (Figure 24).

3.1.4 | Gigantopyramidal neurons

Gigantopyramidal neurons, which were not distinguishable in the mongoose, rabbit, rat, or wallaby, were the deepest neurons traced, with an average soma depth of $1,452 \pm 23 \mu\text{m}$. Of the three neuron types examined, these neurons exhibited the greatest morphological variation across species. Somata varied considerably in shape and size. They tended to be more rounded in several of the feliforms (caracal: Figure 8a–e; clouded leopard: Figure 9i, j; lion: Figure 10a–c) and caniforms (domestic dog: Figure 13a–c) than in other species. They were also unusually large in feliforms (Figure 23u; Table 5)—for example, the average gigantopyramidal neuron soma size in the feliform species examined was $2,847 \mu\text{m}^2$, compared to $987 \mu\text{m}^2$ in the primate species. Basilar dendritic systems of gigantopyramidal neurons were highly variable in terms of number and arrangement, both across and within taxonomic groups. Gigantopyramidal neurons possessed an average of 7.01 ± 3.15 primary basilar dendrites (ranging from 3.7 in the clouded leopard to 14.2 in the giraffe), higher than the superficial or deep pyramidal neurons. Even within feliforms, there was considerable

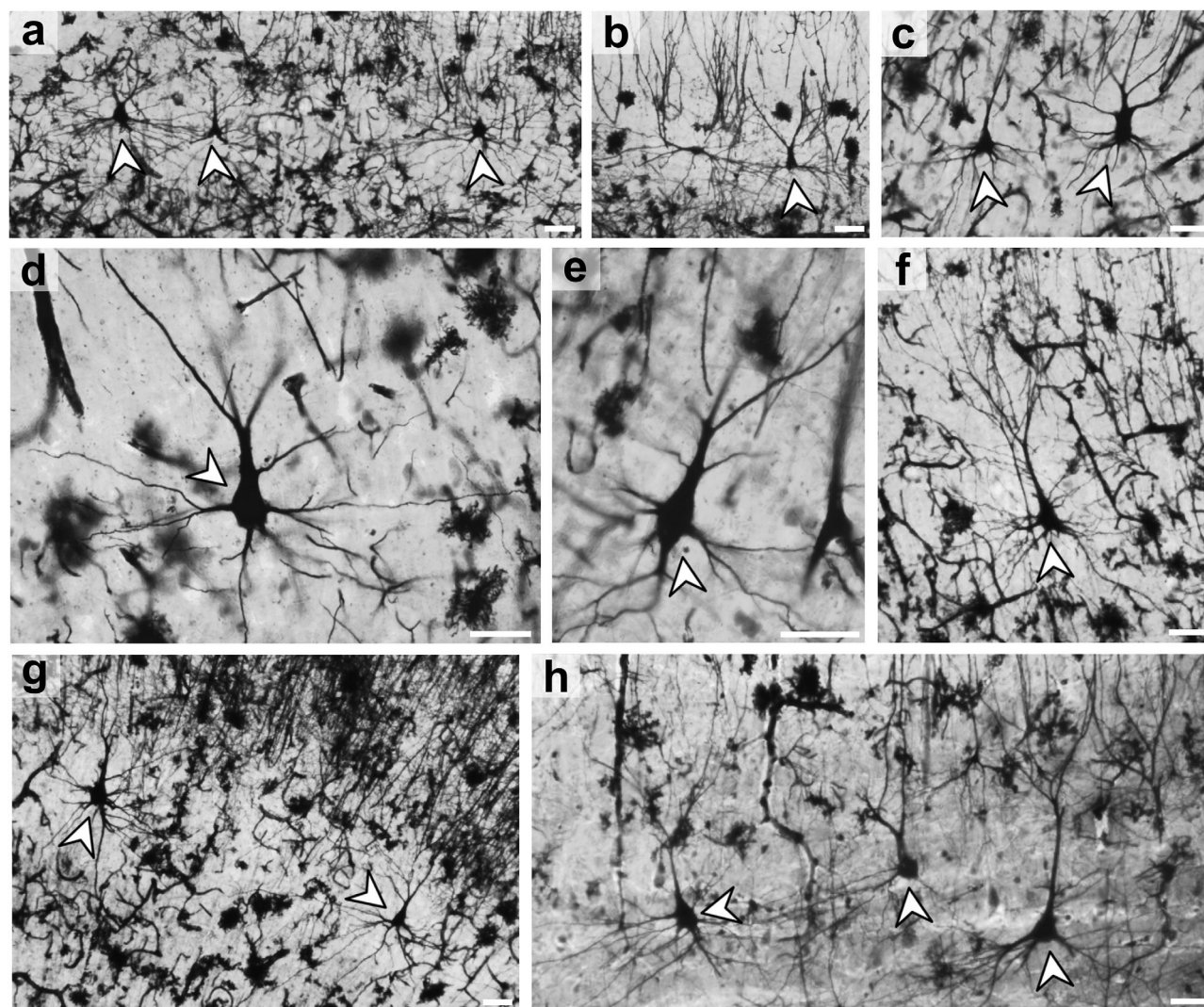


FIGURE 3 Photomicrographs of Golgi-stained gigantopyramidal neurons in feliforms from the following species: caracal (a, b), African lion (c–g), and Siberian tiger (h). Arrowheads indicate somata. Note the numerous circumferential dendrites (a, c, f), and large rounded somata (a, c–h). The tracings of several of these neurons are depicted in Figure 10 such that neuron 3d = tracing 10b, neuron 3e = tracing 10c, and neuron 3f = tracing 10a. Scale bars = 100 μ m

variation as the caracal (9.0) and lion (7.3) exhibited more primary basilar dendrites than did the clouded leopard (3.7) or tiger (3.8).

In terms of morphology, basilar dendritic systems in gigantopyramidal neurons were idiosyncratic, although they were sometimes also severely truncated (e.g., in the clouded leopard and the domestic dog). In some species, circumferential dendrites radiated symmetrically from the entire soma (caracal: Figure 8a, c, e; lion: Figure 10a); in others, basilar dendrites were lateralized and asymmetric (plains zebra: Figure 15a; baboon: Figure 18b); and, in others, basilar dendrites descended primarily from the bottom of the soma (tiger: Figure 9a–c; domestic dog: Figure 13a–c; mountain zebra: Figure 14a–c; kudu: Figure 17j, k), sometimes forming a long taproot (African wild dog: Figure 12b; human: Figure 19b, c; golden lion tamarin: Figure 20b). Apical dendrites, which were truncated in some species (e.g., caracal and clouded leopard), also exhibited considerable variation across taxonomic groups. In some artiodactyls (giraffe: Figure 16a–c; wildebeest: Figure 17a–c), perissodactyls (plains zebra: Figure 15a–c), and feliforms (caracal:

Figure 8b, d; tiger: Figure 9a–c; lion: Figure 10a–c), apical dendrites were characterized by a V-shaped bifurcation. In contrast, apical dendrites in other taxonomic groups often ascended to the pial surface without clear bifurcations (African wild dog: Figure 12a–c; baboon: Figure 18b–d; human: Figure 19a–c; golden lion tamarin: Figure 20a–c).

Quantitatively, gigantopyramidal neurons typically exhibited higher dendritic values than either superficial or deep pyramidal neurons, especially for Vol and MSL (Figure 23c, i; Table 5). This was particularly true for feliforms, where dendritic Vol for gigantopyramidal neurons was 45% higher than the average dendritic Vol in all other taxonomic groups. Within primates, baboon gigantopyramidal neurons exhibited more complex dendritic systems than did other species (Table 5). Spine measures (Figure 23o, r) tended to be lower for gigantopyramidal neurons than for other neuron types, although there was considerable variation across species (Table 5). In the Sholl analysis, gigantopyramidal neurons displayed a pattern similar to that of deep pyramidal neurons, although gigantopyramidal neuron dendrites tended to extend further

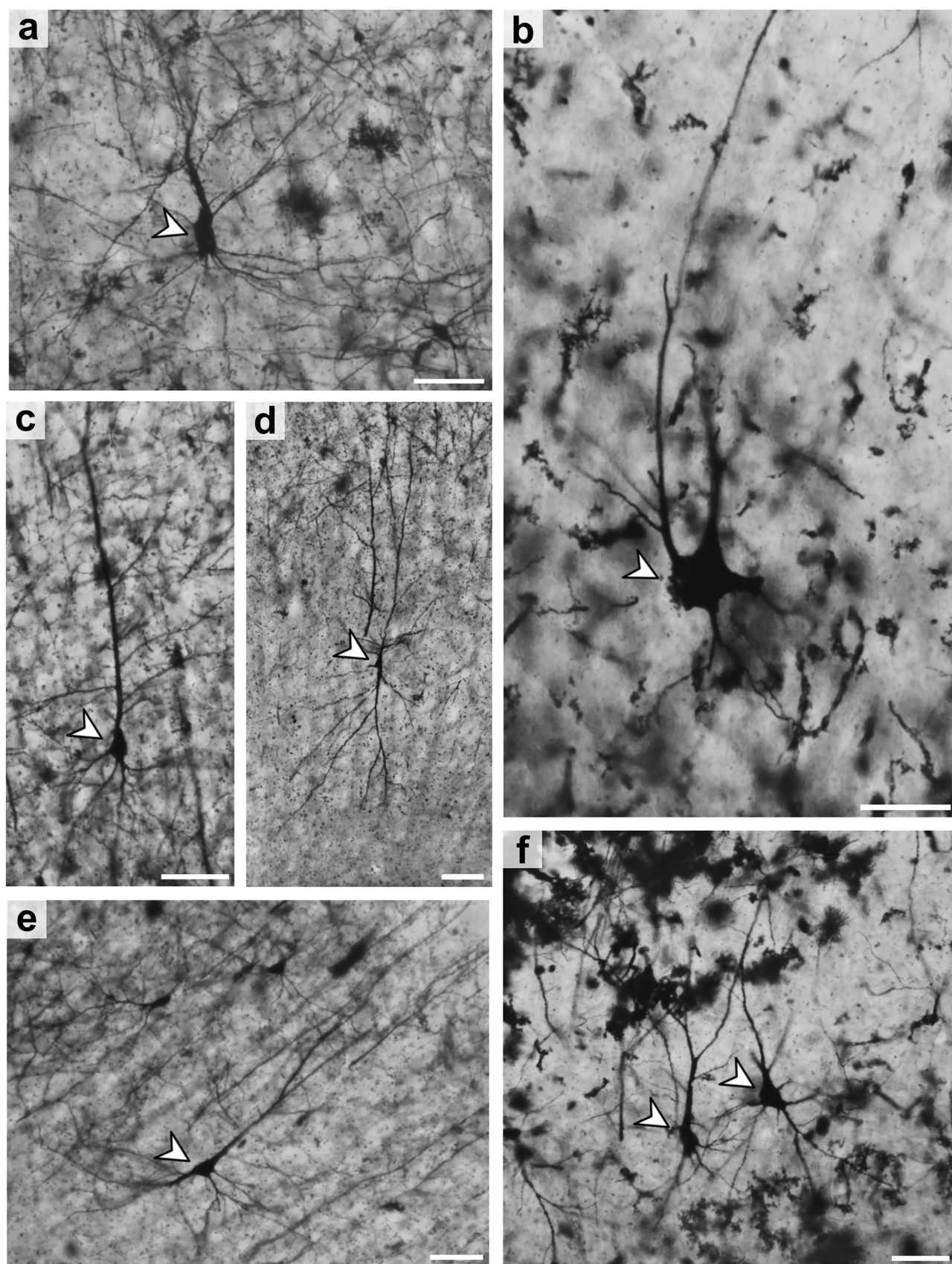


FIGURE 4 Photomicrographs of Golgi-stained gigantopyramidal neurons in caniforms from the following species: African wild dog (a, c–e) and domestic dog (b, f). Arrowheads indicate somata. Note the large rounded somata (b, e, f). The tracings of several of these neurons are depicted in Figures 12 and 13 such that neuron 4b = tracing 13a, neuron 4c = tracing 12a, and neuron 4e = tracing 12c. Scale bars = 100 μ m

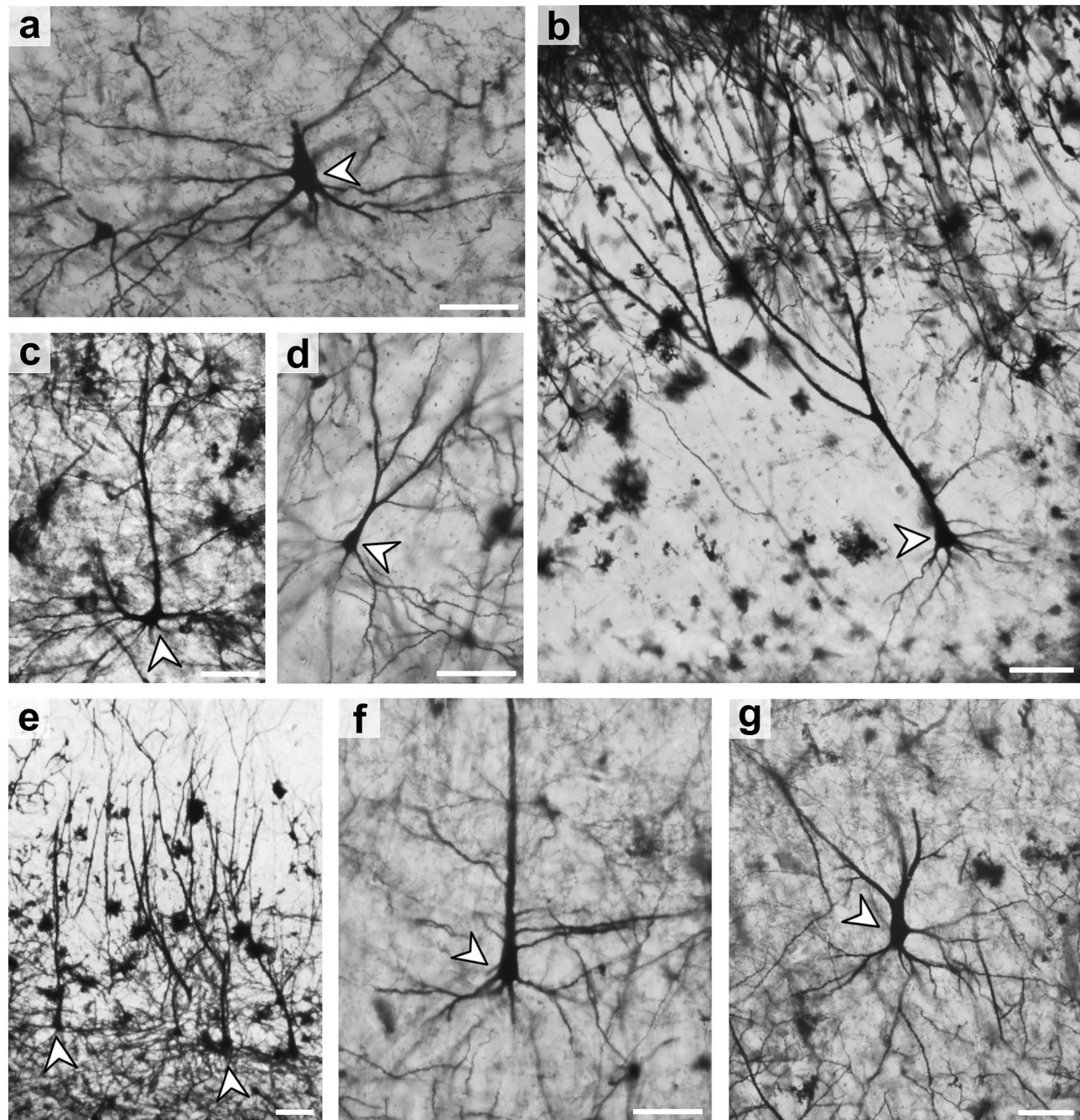


FIGURE 5 Photomicrographs of Golgi-stained gigantopyramidal neurons in perissodactyls from the following species: mountain zebra (a, c, f, g) and plains zebra (b, d, e). Arrowheads indicate somata. Note the V-shaped bifurcations of the apical dendrites (b, d, g). The tracings of several of these neurons are depicted in Figures 14 and 15 such that neuron 5b = tracing 15c, neuron 5c = tracing 14a, and neuron 5f = tracing 14b. Scale bars = 100 μ m

from the soma than did those of deep pyramidal neurons in most taxonomic groups (Figure 24).

3.1.5 | Statistical analyses of neuronal morphology by taxonomic groups

The Kruskal-Wallis test by ranks results rejected the null hypothesis of no differences in dependent measures among neuron types. As indicated in the descriptive measures (Table 5), gigantopyramidal neurons exhibited a significantly greater Vol, $H_{(2,523)} = 251.76$, $p \leq .0001$, TDL,

$H_{(2,523)} = 66.48$, $p \leq .00001$, MSL, $H_{(2,523)} = 108.04$, $p \leq .0001$, and soma size, $H_{(2,523)} = 288.65$, $p \leq .0001$, than did superficial or deep pyramidal neurons. DSC values for gigantopyramidal neurons were greater than those of deep pyramidal neurons, $H_{(2,523)} = 7.48$, $p \leq .0238$, but not superficial pyramidal neurons. For spine measures, DSN values did not differ significantly among neuron types. DSD values for gigantopyramidal neurons were less than the values for superficial and deep pyramidal neurons, $H_{(2,523)} = 29.85$, $p \leq .0001$. These descriptive results thus suggested additional statistical testing was warranted.

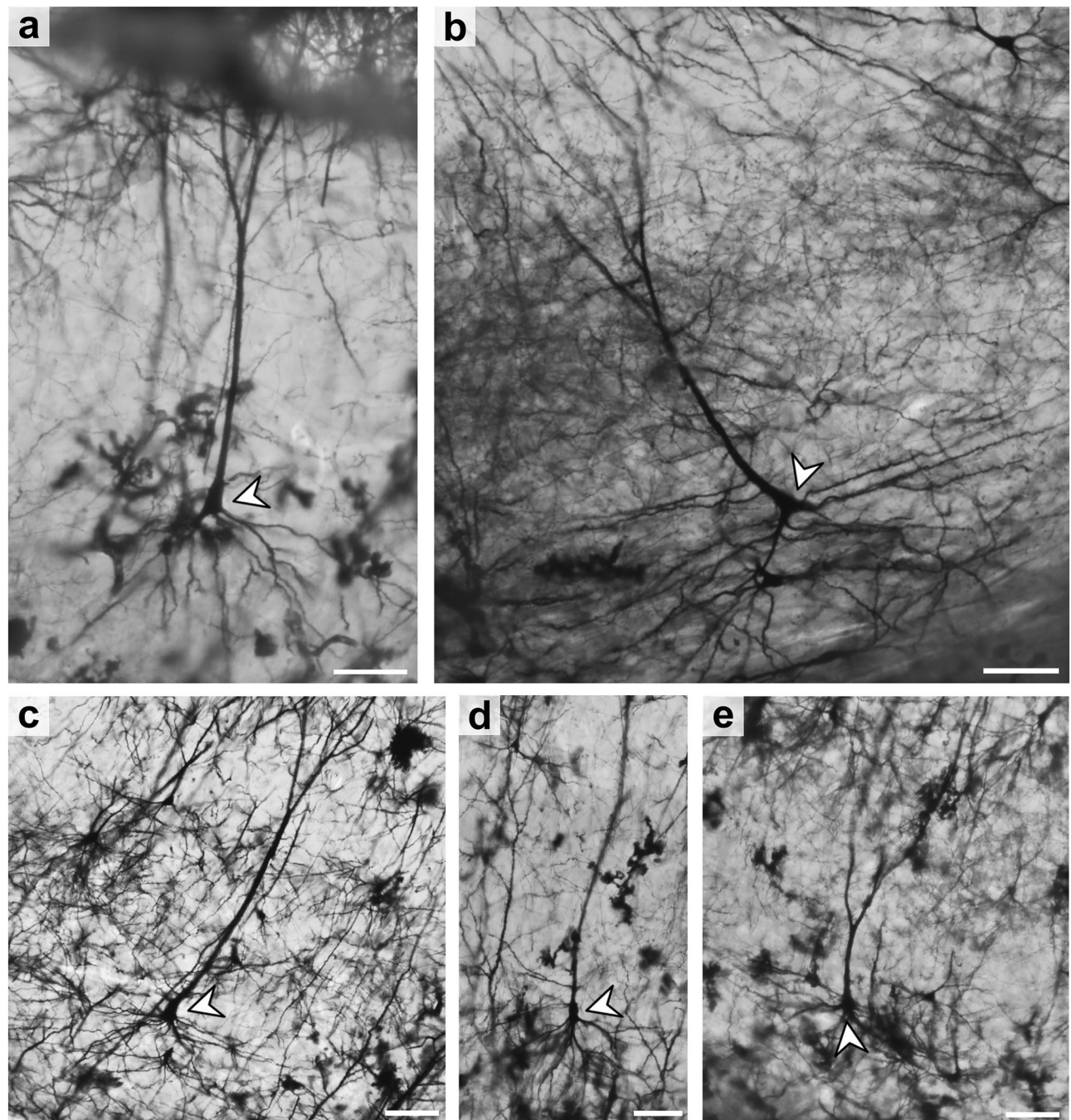


FIGURE 6 Photomicrographs of Golgi-stained gigantopyramidal neurons in artiodactyls from the following species: giraffe (c) and blue wildebeest (a, b, d, e). Arrowheads indicate somata. Note the circumferential dendrites (b) and the relatively slender somata (a–e). The tracing of neuron 6c is depicted in Figure 16c. Scale bars = 100 μ m

In the examination of neuronal attributes among taxonomic groups, both superficial and gigantopyramidal neurons, but not deep pyramidal neurons, clustered at the required overall minimum of 75% accuracy for taxonomic assignment. For superficial pyramidal neurons, artiodactyls, caniforms, feliforms, perissodactyls, and primates could be correctly classified with no errors (Table 6). For these taxonomic groups, classification accuracy ranged from 71% in caniforms to 100% in artiodactyls, with an overall accuracy of 85%. However, superficial pyramidal neurons were poorly discriminated in the diprotodont,

lagomorph, and rodent orders. For superficial pyramidal neurons, all dependent measures were significant differentiators of taxonomic groups: Vol, $F_{(4,215)} = 5.75$, $p \leq .001$; TDL, $F_{(4,215)} = 7.28$, $p \leq .001$; MSL, $F_{(4,215)} = 12.28$, $p \leq .001$; DSC, $F_{(4,215)} = 14.38$, $p \leq .001$; DSN, $F_{(4,215)} = 5.82$, $p \leq .001$; DSD, $F_{(4,215)} = 5.57$, $p \leq .001$; soma size, $F_{(4,215)} = 3.59$, $p \leq .01$.

The cluster analysis for gigantopyramidal neurons was more limited in scope because, as indicated in Table 7, there were no relatively complete gigantopyramidal neurons in some species (e.g., clouded leopard,

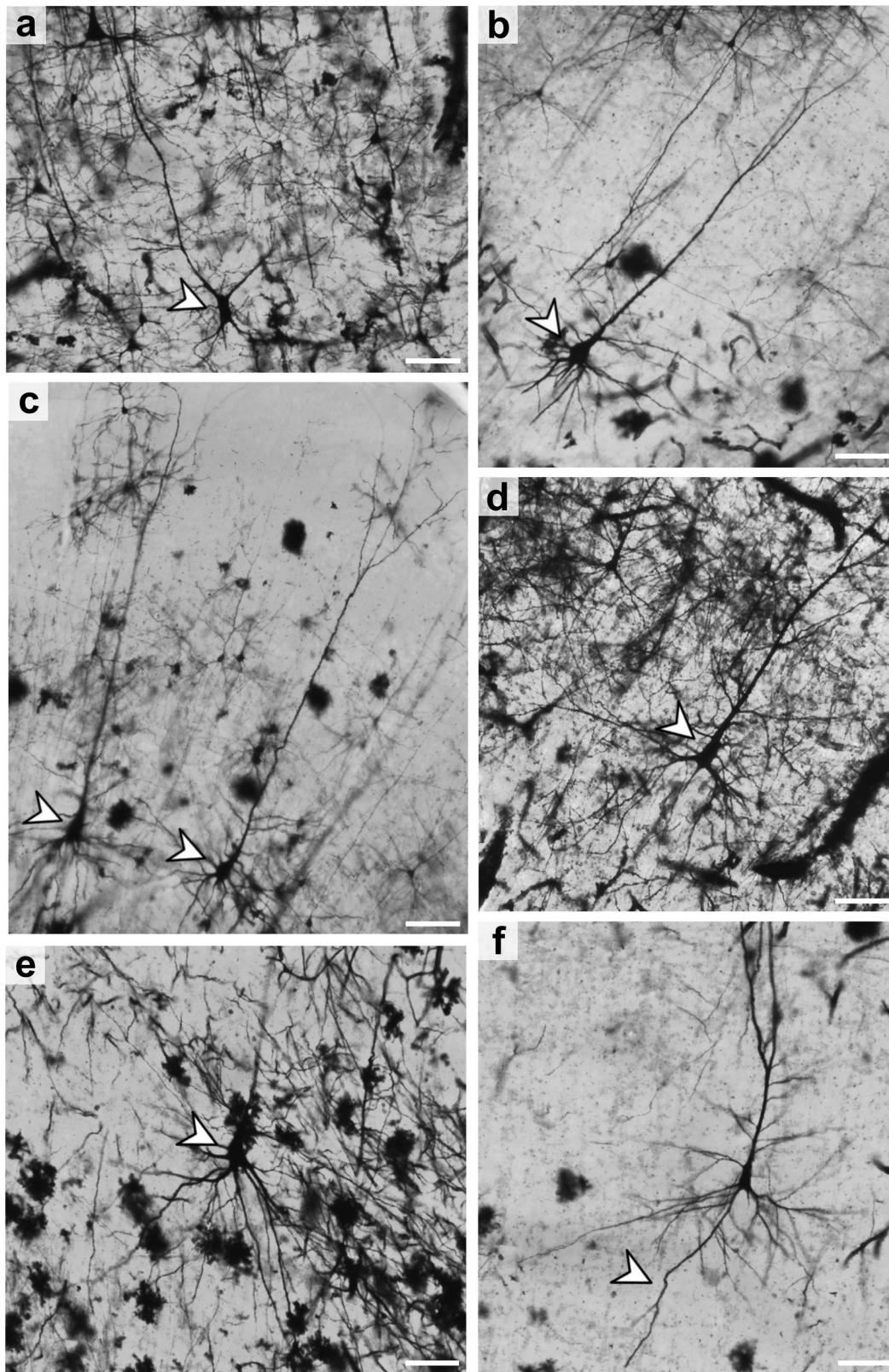


FIGURE 7 Photomicrographs of Golgi-stained gigantopyramidal neurons in several primate species: human (a, d), Golden lion tamarin (b, c), and chacma baboon (e, f). Arrowheads indicate somata, except for (f), where the arrowhead indicates a descending taproot. Note the circumferential dendrites (b, c), and mostly single, ascending apical dendrites (b-d). The tracing of neuron 7f is depicted in Figure 18a. Scale bars = 100 μ m

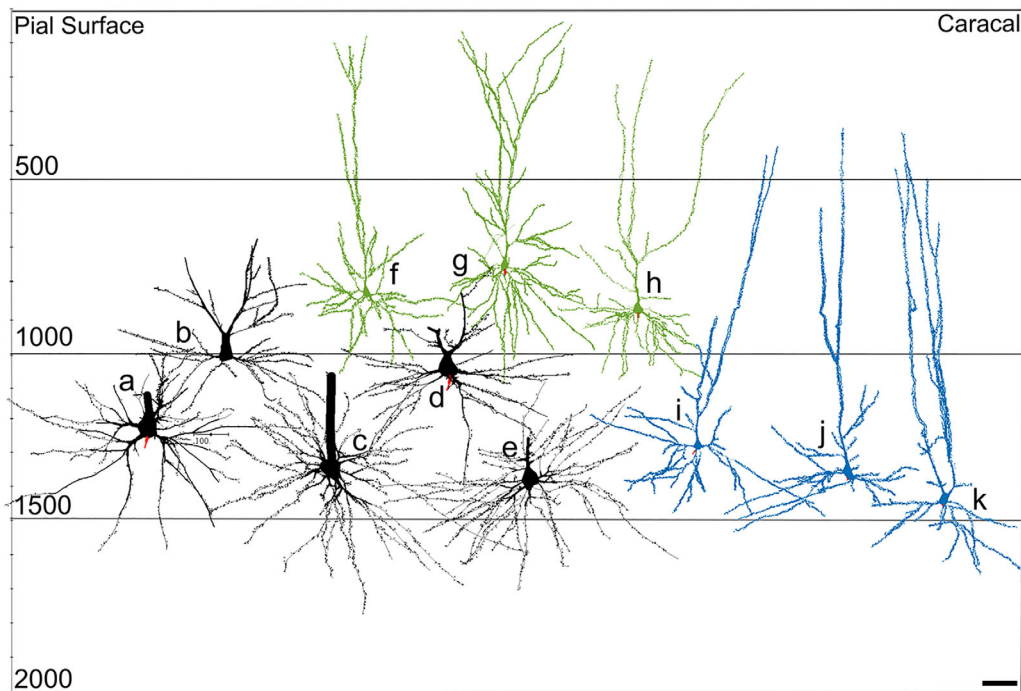


FIGURE 8 Neurolucida tracings of neurons in the primary motor cortex of the caracal indicating relative soma depth from the pial surface (in μm): gigantopyramidal neurons (a–e), superficial pyramidal neurons (f–h), and deep pyramidal neurons (i–k). In gigantopyramidal neurons, note the large, rounded somata, the profuse basilar dendritic skirts, and obviously truncated apical dendrites. Scale bar = 100 μm [Color figure can be viewed at wileyonlinelibrary.com]

mongoose, lemur, rabbit, wallaby, rat). Table 7 indicates that all remaining taxonomic groups examined, except caniforms at 13% and perissodactyls at 71%, were well classified at an overall accuracy of 83%, with a high of 94% correct for feliforms and primates. For gigantopyramidal neurons, all dependent measures were significant differentiators of taxonomic groups: Vol, $F_{(4,117)} = 17.84$, $p \leq .001$; TDL, $F_{(4,117)} = 14.25$, $p \leq .001$; MSL, $F_{(4,117)} = 6.86$, $p \leq .001$; DSC, $F_{(4,117)} = 11.78$, $p \leq .001$; DSN, $F_{(4,117)} = 31.69$, $p \leq .001$; DSD, $F_{(4,117)} = 38.44$, $p \leq .001$; soma size, $F_{(4,117)} = 56.10$, $p \leq .01$.

3.2 | Stereological analysis

Photomicrographs of Nissl-stained gigantopyramidal neurons of all carnivore and primate species examined are provided in Figure 25. We analysed the soma measures of layer V pyramidal and gigantopyramidal neurons in the primary motor cortex of nine primates and 11 carnivores (Table 3). A summary of the resulting bivariate relationships is presented in Table 8.

In primates, the average soma area ranged from 107.7 μm^2 in the golden lion tamarin to 205.8 μm^2 in the gibbon for layer V pyramidal neurons, and from 183.1 μm^2 in the golden lion tamarin to 352.6 μm^2 in the vervet monkey for gigantopyramidal neurons (Table 3), indicating that the average soma area of gigantopyramidal neurons was 1.64 times larger than that of layer V pyramidal neurons. In terms of soma volume, the average ranged from 909.3 μm^3 in the golden lion tamarin to 2,558.9 μm^3 in the gibbon for layer V pyramidal neurons, and from 2,171.0 μm^3 in the golden lion tamarin to

5,636.5 μm^3 in the vervet monkey for gigantopyramidal neurons (Table 3), indicating that the average soma volume of gigantopyramidal neurons was 2.3 times larger than that of layer V pyramidal neurons. No statistically significant relationships were observed in soma measures for either layer V pyramidal or gigantopyramidal neurons in the primate species examined (Table 8); thus, no clear increases in soma size occurred with increased brain or body mass in primates (Figure 26).

In non-P carnivores, the average soma area ranged from 187.5 μm^2 in the mongoose to 319.1 μm^2 in the seal for layer V pyramidal neurons, and from 330.3 μm^2 in the mongoose to 872.8 μm^2 in the otter for gigantopyramidal neurons (Table 3), indicating that the average soma area of gigantopyramidal neurons was 2.4 times larger than that of layer V pyramidal neurons. For soma volume, the average ranged from 2,099.2 μm^3 in the mongoose to 4,667.7 μm^3 in the seal for layer V pyramidal neurons, and from 5,297.7 μm^3 in the mongoose to 20,560.2 μm^3 in the otter for gigantopyramidal neurons (Table 3), indicating that the average soma volume of gigantopyramidal neurons was 3.9 times larger than that of layer V pyramidal neurons. For layer V pyramidal neurons in non-P carnivores, statistically significant positive relationships were observed between the soma measures and both brain and body mass (Table 8). For gigantopyramidal neurons in non-P carnivores, however, there was no statistically significant relationships between soma measures and brain or body mass (Table 8). In general, soma measures for layer V pyramidal and gigantopyramidal neurons were significantly larger in non-P carnivores than in primates (Figure 26), $F_{(11,90)} = 45.98$, $p \leq .0001$.

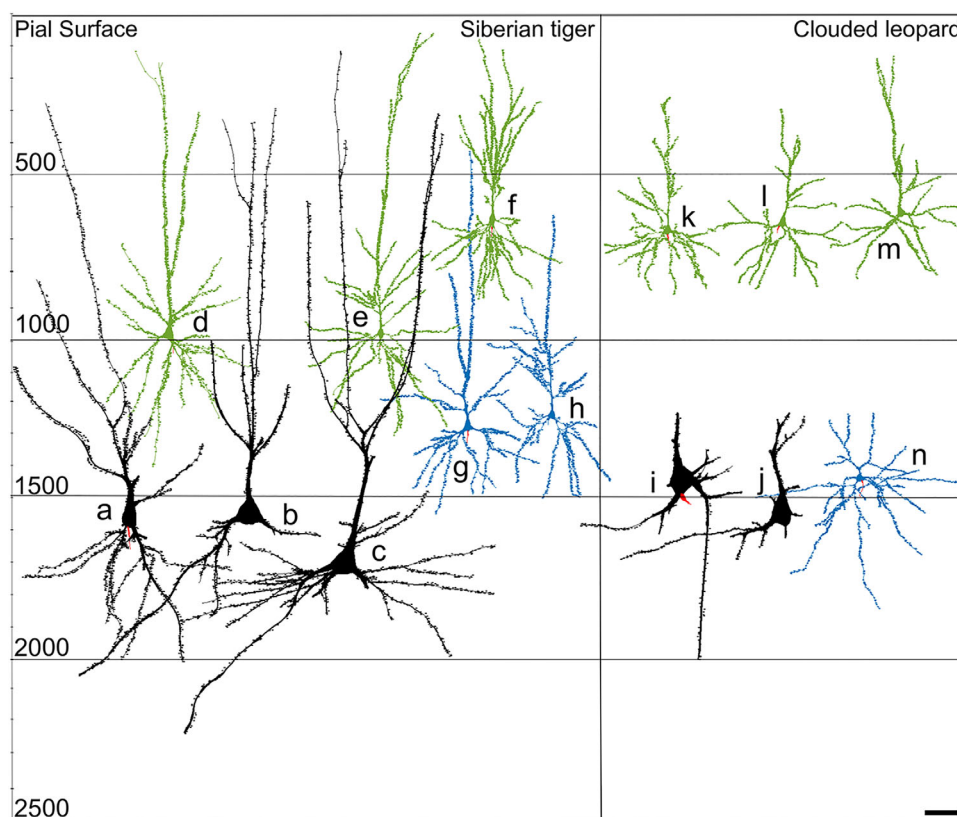


FIGURE 9 Neurolucida tracings of neurons in the primary motor cortex of the Siberian tiger (left) and clouded leopard (right) indicating relative soma depth from the pial surface (in μm): gigantopyramidal neurons (a–c, i, j), superficial pyramidal neurons (d–f; k–m), and deep pyramidal neurons (g, h, n). Note the very large somata of gigantopyramidal neurons in both species, the descending taproot dendrites (b, c), and the incomplete tracings of gigantopyramidal neurons in the clouded leopard. Adapted from figures 6 and 7 of Johnson et al. (2016). Scale bar = 100 μm [Color figure can be viewed at wileyonlinelibrary.com]

In the three *Panthera* species, the average soma area ranged from 225.9 μm^2 in the Amur leopard to 268.9 μm^2 in the lion for layer V pyramidal neurons, and from 1,040.0 μm^2 in the Siberian tiger to 1,416.2 μm^2 in the lion for gigantopyramidal neurons (Table 3), indicating that the average soma area of gigantopyramidal neurons was 5.1 times larger than that of layer V pyramidal neurons. For soma volume, the average ranged from 2,775.7 μm^3 in Amur leopard to 3,793.6 μm^3 in the lion for layer V pyramidal neurons, and from 32,289.6 μm^3 in the Siberian tiger to 46,173.6 μm^3 in the lion for gigantopyramidal neurons (Table 3), indicating that the average soma volume of gigantopyramidal neurons was 12.26 times larger than that of layer V pyramidal neurons. Although the soma measures of layer V pyramidal neurons in the *Panthera* species did not differ significantly from those observed for non-P carnivores (Table 8), the soma measures of the gigantopyramidal neurons were significantly larger than those of the non-P carnivores, $F_{(4,20)} = 63.32$, $p \leq .0001$ (Figures 25–26).

4 | DISCUSSION

The current results supplement a growing comparative database of neuronal morphology (Ascoli, Donohue, & Halavi, 2007; Bianchi et al., 2013; Butti et al., 2015; Jacobs et al., 2011, 2014; Jacobs,

Harland, et al., 2015; Johnson et al., 2016; Reyes, Harland, Reep, Sherwood, & Jacobs, 2016) by documenting three types of neocortical neurons across a diverse sampling of mammals, and comparing the soma size of layer V pyramidal neurons and gigantopyramidal neurons between selected carnivore and primate species. The present study constitutes the first comparative exploration of quantitative dendritic morphology in gigantopyramidal neurons. The morphological findings in superficial and deep pyramidal neurons are generally consistent with what has been observed previously in placental mammals (carnivores: Fox, Inman, & Himwich, 1966; Hübener, Schwarz, & Bolz, 1990; Matsubara, Chase, & Thejomayen, 1996; Srivastava, Singh, & Chauhan, 2013; Winer, 1984; Yamamoto, Samejima, & Oka, 1987; cetartiodactyls: Butti et al., 2014, 2015; primates: Bianchi et al., 2013; de Lima, Voigt, & Morrison, 1990; Elston, Benavides-Piccione, & DeFelipe, 2001; Feldman, 1984; Ghosh & Porter, 1988; Meyer, 1987; Mohan et al., 2015; Valverde, 1986). Statistically, the specific constellation of somatodendritic measures for superficial and gigantopyramidal neurons could be used to accurately distinguish taxonomic groups. The present morphological findings also indicated that gigantopyramidal neurons were significantly larger and more morphologically variable than either superficial or deep pyramidal neurons across species. Stereological analyses complemented the morphological findings by indicating that (1) soma

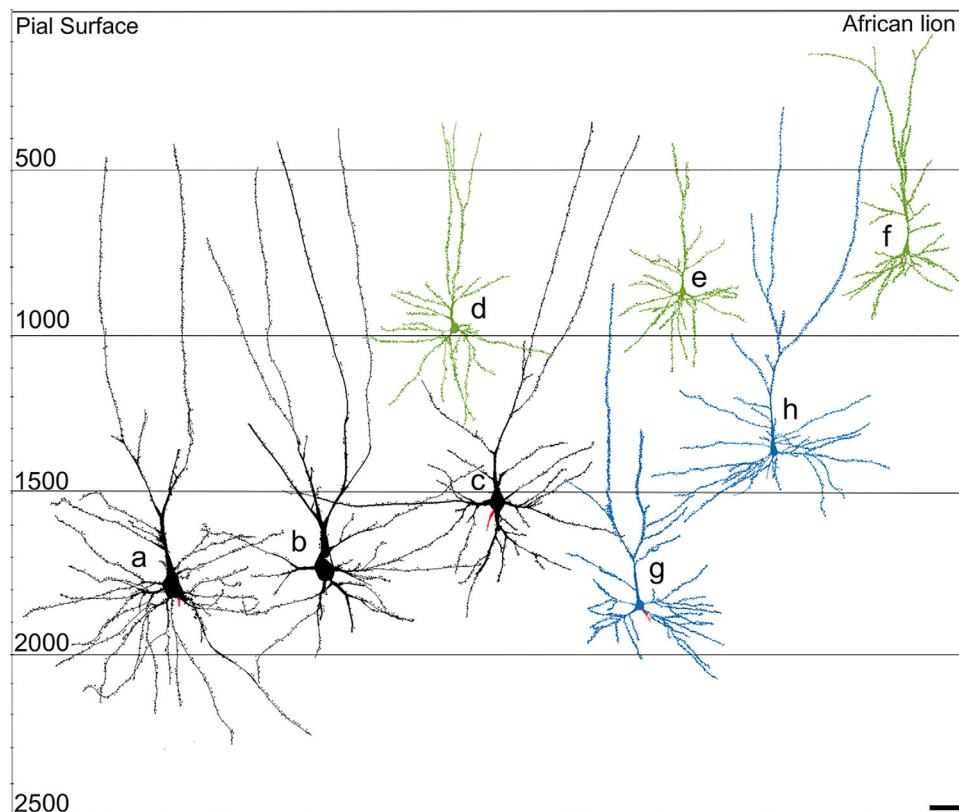


FIGURE 10 Neurolucida tracings of neurons in the primary motor cortex of the African lion indicating relative soma depth from the pial surface (in μm): gigantopyramidal neurons (a–c), superficial pyramidal neurons (d–f), and deep pyramidal neurons (g, h). Note the relatively large, somewhat rounded somata and bifurcating apical dendrites of the gigantopyramidal neurons. Scale bar = 100 μm [Color figure can be viewed at wileyonlinelibrary.com]

measures in gigantopyramidal neurons were larger than in layer V pyramidal neurons for both carnivores and primates, (2) soma measures in gigantopyramidal neurons were larger in carnivores than in primates, and (3) within carnivores, gigantopyramidal neuron somata

were exceptionally large in the three *Panthera* species. Below, following a brief consideration of methodological issues in the morphological analysis, we explore the findings in more detail with a focus on gigantopyramidal neurons in each taxonomic group.

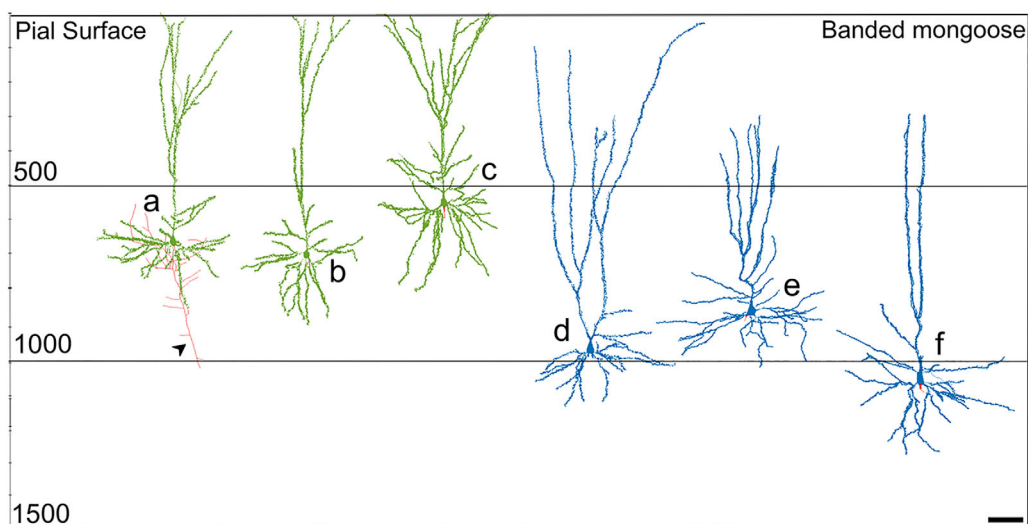


FIGURE 11 Neurolucida tracings of neurons in the primary motor cortex of the banded mongoose indicating relative soma depth from the pial surface (in μm): superficial pyramidal neurons (a–c) and deep pyramidal neurons (d–f). No gigantopyramidal neurons were distinguishable in the Golgi-impregnation for this species. Arrowhead indicates the axon from neuron a. Scale bar = 100 μm [Color figure can be viewed at wileyonlinelibrary.com]

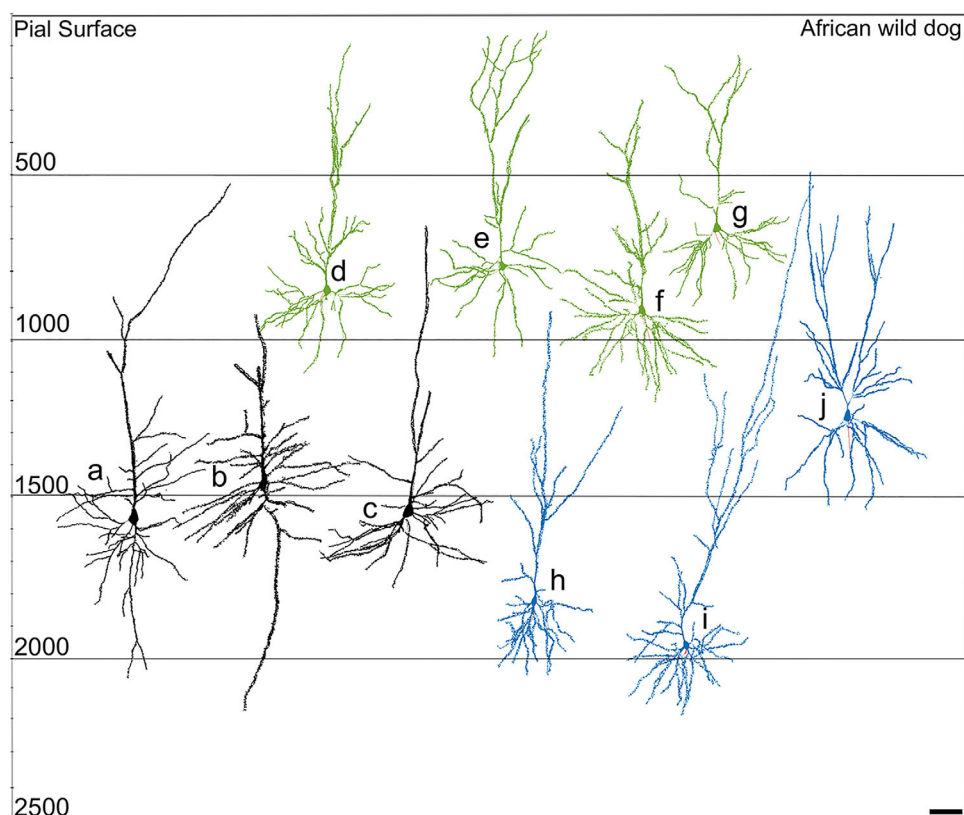


FIGURE 12 Neurolucida tracings of neurons in the primary motor cortex of the African wild dog indicating relative soma depth from the pial surface (in μm): gigantopyramidal neurons (a–c), superficial pyramidal neurons (d–g), and deep pyramidal neurons (h–j). Note the relatively rounded somata in the gigantopyramidal neurons, and the descending, taproot-like basilar dendrite (b). Scale bar = $100\ \mu\text{m}$ [Color figure can be viewed at wileyonlinelibrary.com]

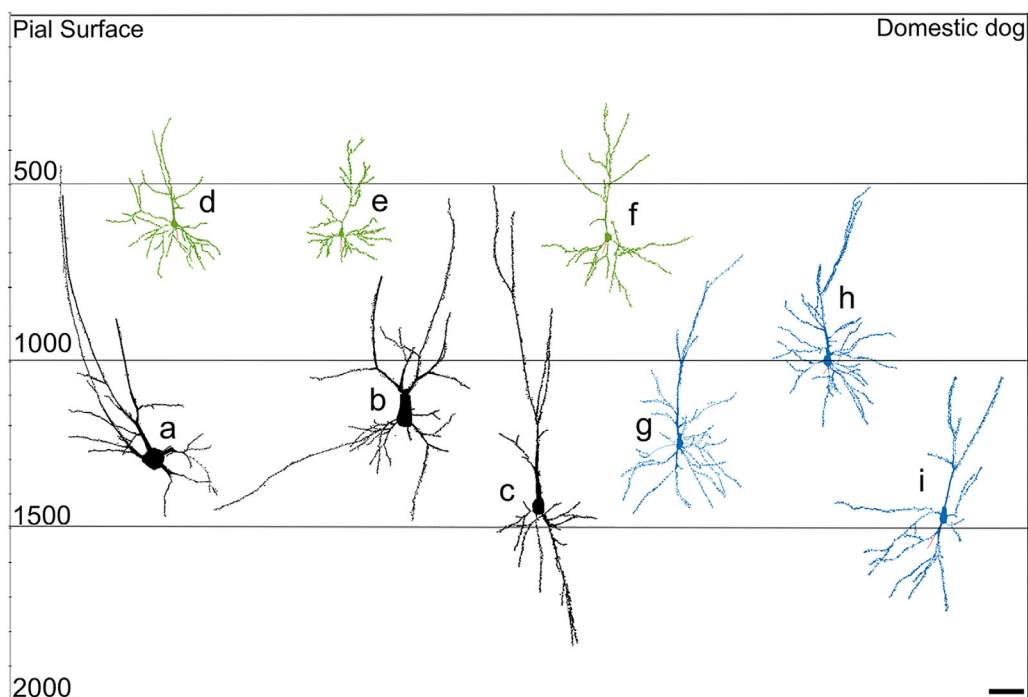


FIGURE 13 Neurolucida tracings of neurons in the primary motor cortex of the domestic dog indicating relative soma depth from the pial surface (in μm): gigantopyramidal neurons (a–c), superficial pyramidal neurons (d–f), and deep pyramidal neurons (g–i). Note the large and rounded somata in the rather incomplete tracings of gigantopyramidal neurons. Scale bar = $100\ \mu\text{m}$ [Color figure can be viewed at wileyonlinelibrary.com]

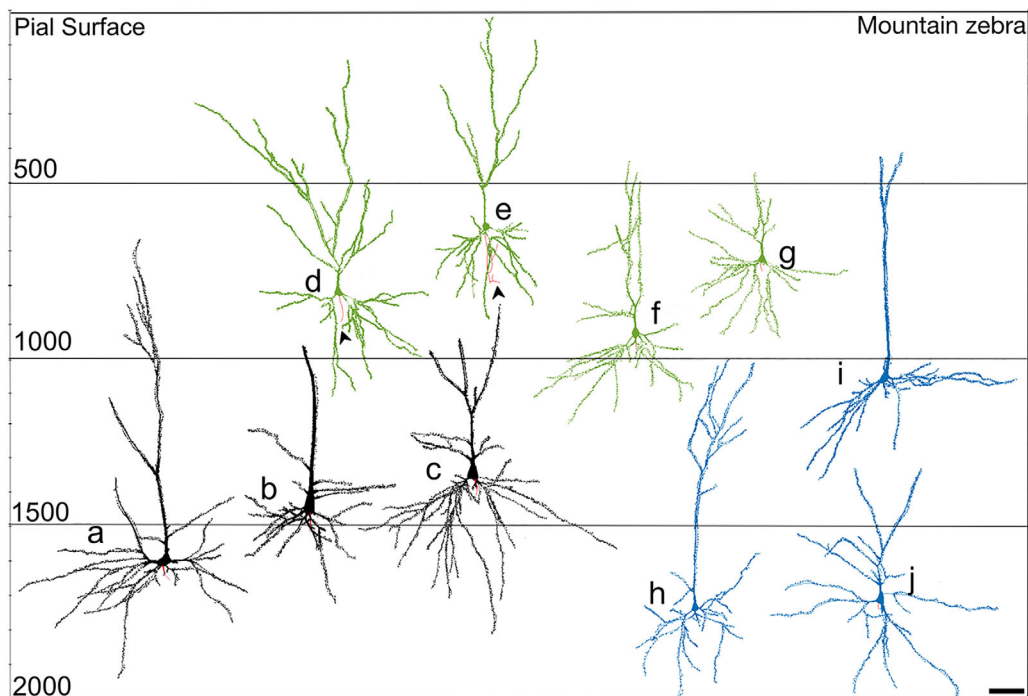


FIGURE 14 Neurolucida tracings of neurons in the primary motor cortex of the mountain zebra indicating relative soma depth from the pial surface (in μm): gigantopyramidal neurons (a–c), superficial pyramidal neurons (d–g), and deep pyramidal neurons (h–j). Note the bifurcating apical dendrite (a). Arrowheads indicate axons from neurons (d) and (e). Scale bar = 100 μm [Color figure can be viewed at wileyonlinelibrary.com]

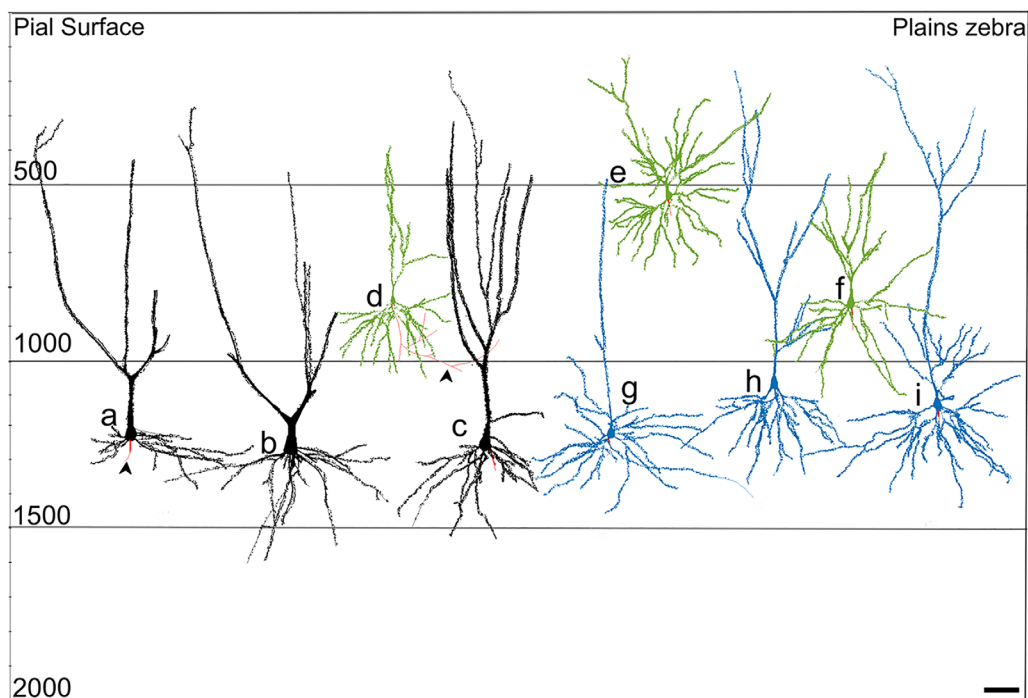


FIGURE 15 Neurolucida tracings of neurons in the primary motor cortex of the plains zebra indicating relative soma depth from the pial surface (in μm): gigantopyramidal neurons (a–c), superficial pyramidal neurons (d–f), and deep pyramidal neurons (g–i). Note the relatively wide V-shaped apical dendritic bifurcations in the gigantopyramidal neurons, which are much more distinct than in the mountain zebra (Figure 14). Arrowheads indicate axons from neurons (a) and (d). Scale bar = 100 μm [Color figure can be viewed at wileyonlinelibrary.com]

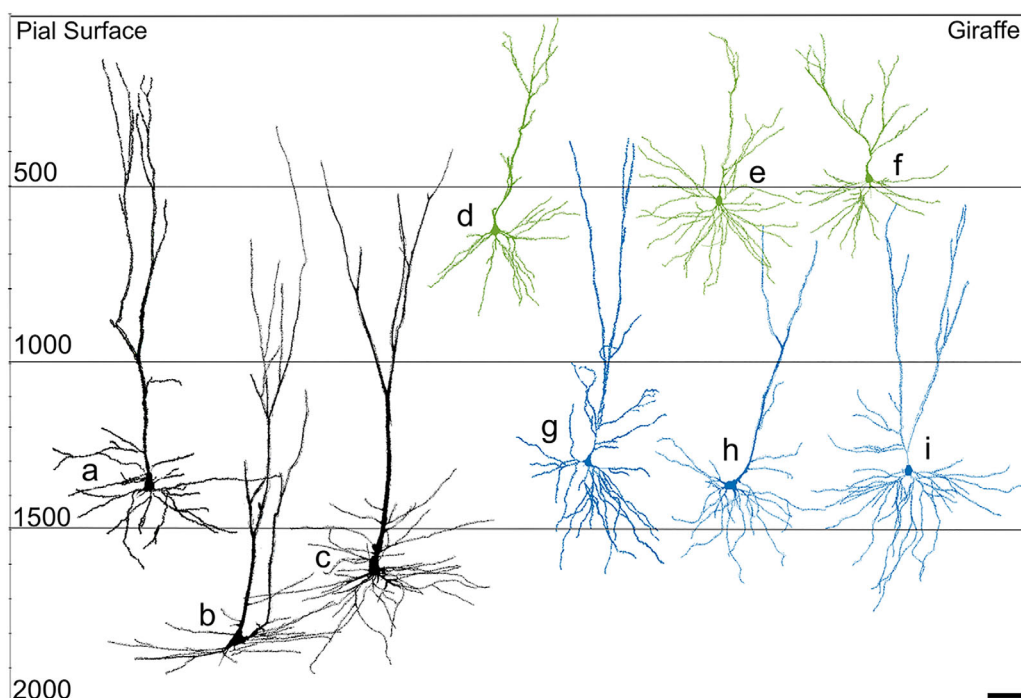


FIGURE 16 Neurolucida tracings of neurons in the primary motor cortex of the giraffe indicating relative soma depth from the pial surface (in μm): gigantopyramidal neurons (a–c), superficial pyramidal neurons (d–f), and deep pyramidal neurons (g–i). Note the narrow V-shaped bifurcations in the apical dendrites in both gigantopyramidal and deep pyramidal neurons. Adapted from figure 6 of Jacobs et al. (2015). Scale bar = 100 μm [Color figure can be viewed at wileyonlinelibrary.com]

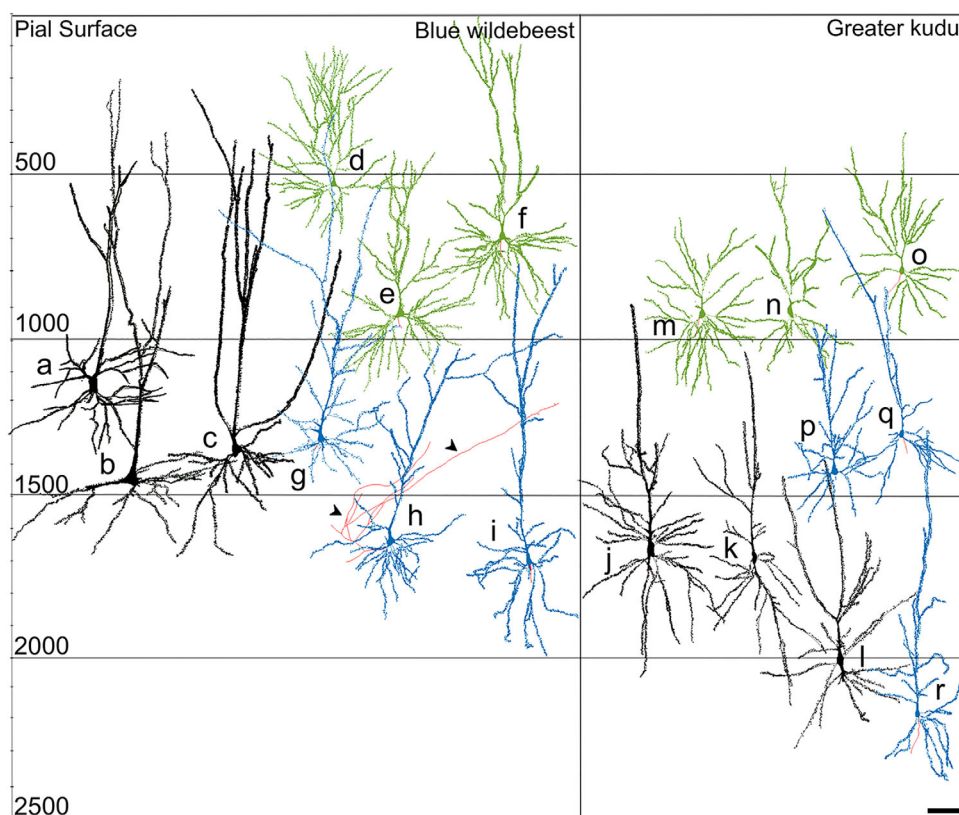


FIGURE 17 Neurolucida tracings of neurons in the primary motor cortex of the blue wildebeest (left) and the greater kudu (right) indicating relative soma depth from the pial surface (in μm): gigantopyramidal neurons (a–c; j–l), superficial pyramidal neurons (d–f; m–o), and deep pyramidal neurons (g–i; p–r). Note the V-shaped bifurcations in the apical dendrites of gigantopyramidal neurons in the blue wildebeest, which are much more distinct than in the greater kudu. Arrowheads indicate the axon from neuron (h). Scale bar = 100 μm [Color figure can be viewed at wileyonlinelibrary.com]

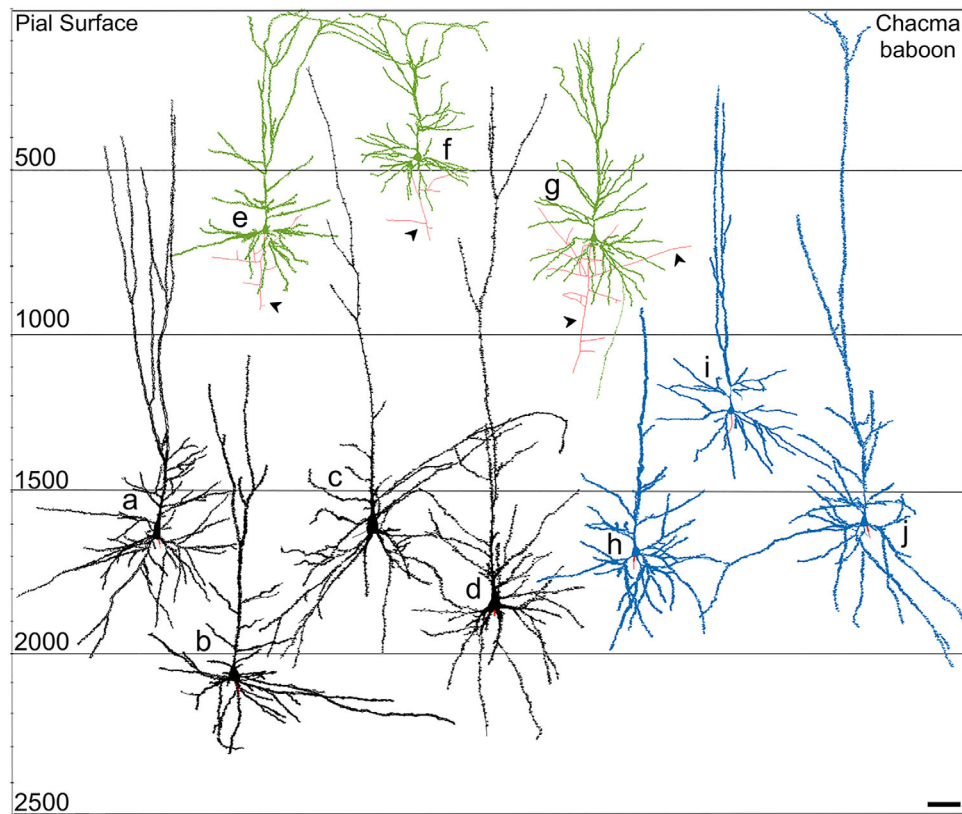


FIGURE 18 Neurolucida tracings of neurons in the primary motor cortex of the chacma baboon indicating relative soma depth from the pial surface (in μm): gigantopyramidal neurons (a–d), superficial pyramidal neurons (e–g), and deep pyramidal neurons (h–j). Note the vertically aligned, mostly singular apical dendrites and complex basilar dendritic arrays of the gigantopyramidal neurons. Arrowheads indicate axons from neurons (e), (f), and (g). Scale bar = 100 μm [Color figure can be viewed at wileyonlinelibrary.com]

4.1 | Methodological considerations of morphological analysis

In accordance with the well-documented limitations of quantitative Golgi investigations (Braak & Braak, 1985; Jacobs et al., 1997, 2011, 2014; Jacobs, Harland, et al., 2015; Jacobs, Lee, et al., 2015; Williams, Ferrante, & Caviness, 1978), the present study is constrained by tissue fixation/storage issues, dendritic truncation in sectioning, and the small sample size of individual specimens and reconstructed neurons. The type of fixation (i.e., perfusion vs. immersion) is known to affect the quality of Golgi impregnations, as is the duration that tissue is stored prior to staining (de Ruiter, 1983; Friedland, Los, & Ryugo, 2006; Johnson et al., 2016). These issues have particularly negative effects on spine estimates (Anderson et al., 2009; Butti et al., 2014; Johnson et al., 2016; Morest & Morest, 1966). Although most tissue in the present study was perfused and stored in sodium azide prior to staining to prevent overfixation, there was still considerable variation in the quality of staining—especially for spines—across species. As such, we have used the spine measures cautiously in our descriptive interpretation of differences across taxonomic groups. Next, as has been noted previously (Jacobs et al., 1997, 2011), larger neurons are more attenuated by tissue sectioning. Thus, our dendritic measures of gigantopyramidal neurons, in particular, are underestimations of their actual morphological complexity. Finally, we had only one or two representatives for

each species in the present sample, and a small number of reconstructions for each type of neuron within each species, which limited potential statistical analyses. For this reason, we inferentially examined differences across taxonomic groups rather than across individual species.

Specific to the present study, the variable distribution exhibited by gigantopyramidal neurons within a single species (Rivara et al., 2003) is further compounded here by the inclusion of multiple species that have not yet been extensively studied for M1 somatotopic representation. The exact electrophysiological properties of neurons in M1 in general, and the hand-forepaw region in particular, remain less clear across many of these species than for well-researched laboratory animals (e.g., macaque; Soares et al., 2017). As such, the gigantopyramidal neurons in the present sample may not represent a homogenous, physiologically equivalent population, which induces additional variability. Although this issue seems unavoidable in such a broad, comparative framework, the availability of well-preserved postmortem brain tissue from such rare species presented a unique opportunity.

4.2 | Carnivores: caniforms and feliforms

Caniforms and feliforms, the two suborders of carnivores (Eizirik et al., 2010; Nyakatura & Bininda-Emonds, 2012), have long been associated with research on gigantopyramidal neurons (Betz, 1874). The

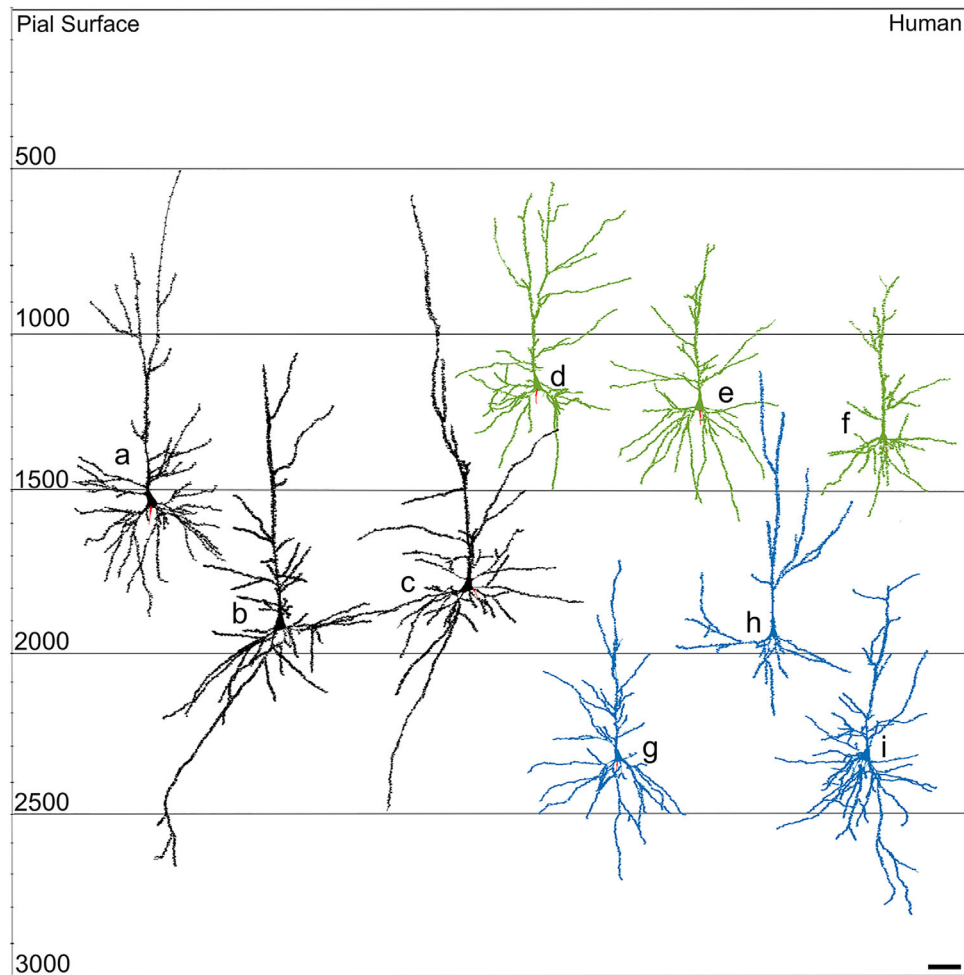


FIGURE 19 Neurolucida tracings of neurons in the primary motor cortex of the human indicating relative soma depth from the pial surface (in μm): gigantopyramidal neurons (a–c), superficial pyramidal neurons (d–f), and deep pyramidal neurons (g–i). In the gigantopyramidal neurons, note the vertically aligned singular apical dendrites, complex basilar dendritic skirts, and descending taproot dendrites (b, c). Scale bar = 100 μm [Color figure can be viewed at wileyonlinelibrary.com]

domestic cat, in particular, has traditionally been the focal animal for both anatomical and physiological studies of gigantopyramidal neurons (Crawford & Curtis, 1966; Hassler & Muhs-Clement, 1964; Kaiserman-Abramof & Peters, 1972; Lewis, 1878; Phillips, 1956; Ramon-Moliner, 1961). In the present morphological analysis, caniform gigantopyramidal somata appeared relatively large, second only to those in feliforms, confirming Brodmann's (1909) observation. There were also morphological differences in gigantopyramidal neurons between the two caniform species, with gigantopyramidal neurons in the domestic dog exhibiting more morphological variability than those in the African wild dog, whose gigantopyramidal neurons tended to resemble those in the mountain zebra. Apart from depictions of Golgi-stained pyramidal neurons in Mervis (1978), which do appear similar to the current superficial and deep caniform neurons, the only morphological comparison point in the literature appears to be the Barasa's (1960) drawings in the domestic dog. The deeper pyramidal neurons in Barasa's Figure 5 are more complete, but do resemble the gigantopyramidal neurons observed here in the domestic dog more than they resemble those in the African wild dog.

Morphologically, feliform gigantopyramidal neurons tended to be more dendritically complex than caniform gigantopyramidal neurons. Within feliforms, tiger and clouded leopard gigantopyramidal neurons had more conical-shaped somata and fewer basilar dendrites compared to the caracal and lion, which exhibited more rounded somata surrounded (especially in the caracal) by a dense basilar dendritic array. However, caracal, tiger, and lion gigantopyramidal neurons expressed similar, V-shaped apical dendrites, which have not only been documented frequently in the domestic cat (fig. 4 in Barasa, 1960; Kaiserman-Abramof & Peters, 1972), but also resemble those observed in some cetartiodactyls (giraffe: Jacobs, Harland, et al., 2015; humpback whale: Butti et al., 2015; wildebeest: current study) and perissodactyls (horse: Barasa, 1960; plains zebra: current study). Some large, layer V domestic cat neurons in the literature resemble those in the tiger and clouded leopard (fig. 8 in Ghosh, Fyffe, & Porter, 1988; fig. 1 in Kaiserman-Abramof & Peters, 1972) whereas others resemble those in the caracal and lion (figs. 5–6 in Deschènes et al., 1979; figs. 9–10 in Ghosh et al., 1988) because of the dense basilar skirt. Large layer V pyramidal neurons in the motor cortex of the domestic cat with a similar

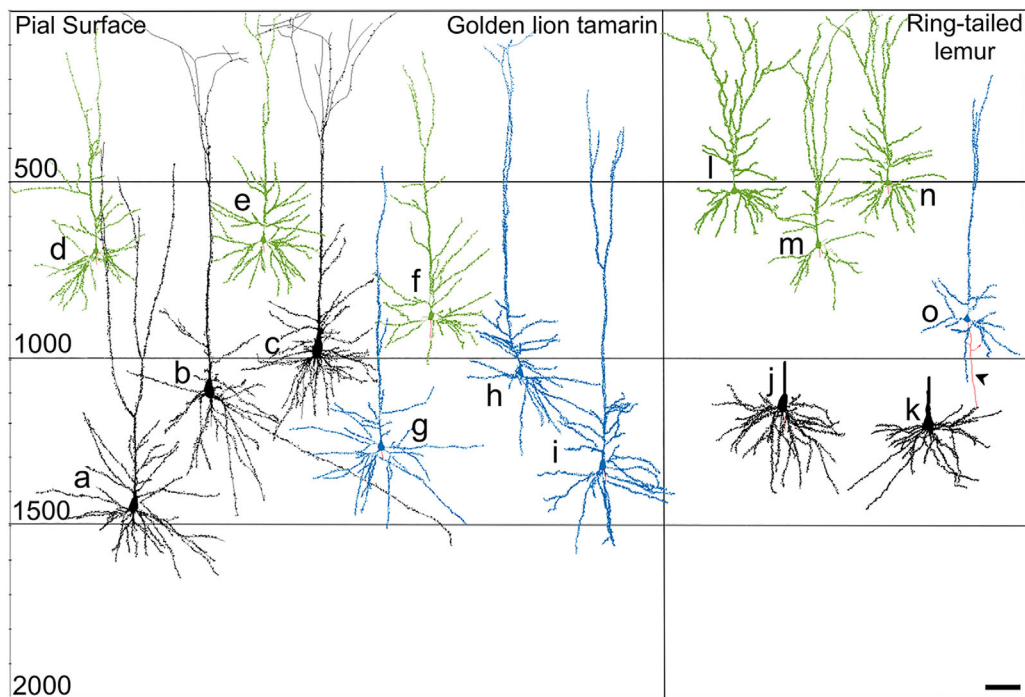


FIGURE 20 Neurolucida tracings of neurons in the primary motor cortex of the golden lion tamarin (left) and the ring-tailed lemur (right) indicating relative soma depth from the pial surface (in μm): gigantopyramidal neurons (a–c; j, k), superficial pyramidal neurons (d–f; l–n), and deep pyramidal neurons (g–i; o). In the tamarin gigantopyramidal neurons, note the vertically aligned, mostly singular apical dendrites and very complex basilar dendritic arrays. In the lemur gigantopyramidal neurons, basilar skirts are complex, but the apical dendrites are truncated. Arrowhead indicates the axon from neuron (o). Scale bar = 100 μm [Color figure can be viewed at wileyonlinelibrary.com]

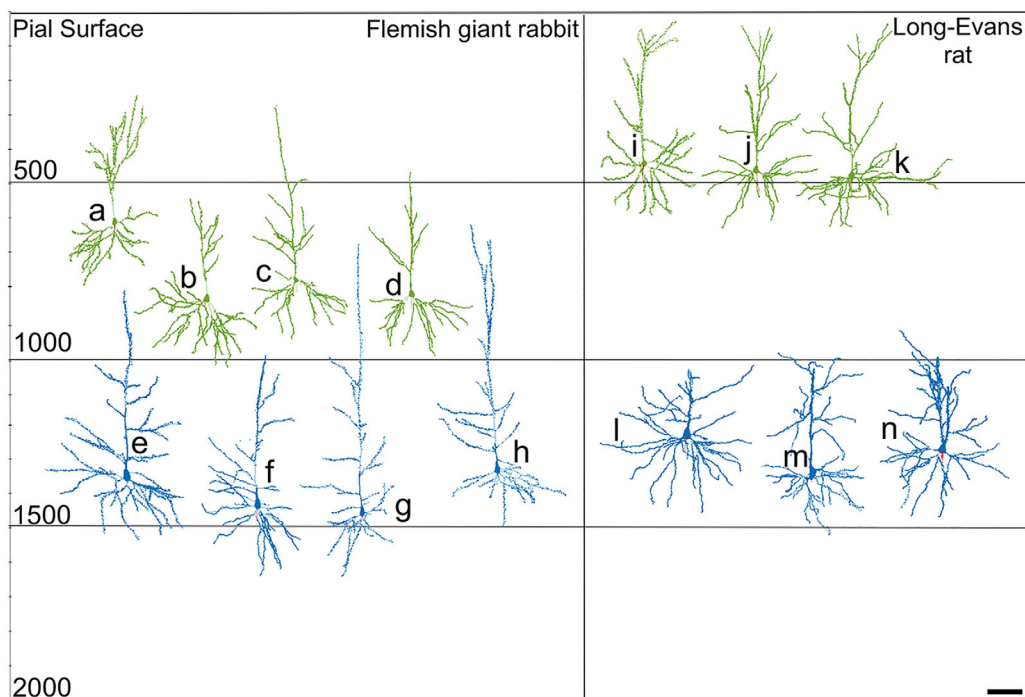


FIGURE 21 Neurolucida tracings of neurons in the primary motor cortex of the Flemish giant rabbit (left) and the Long-Evans rat (right) indicating relative soma depth from the pial surface (in μm): superficial pyramidal neurons (a–d; i–k), and deep pyramidal neurons (e–h; l–n). Note the singular, vertically aligned apical dendrites in both neuron types for both species. No gigantopyramidal neurons were distinguishable in the Golgi-impregnations for these species. Scale bar = 100 μm [Color figure can be viewed at wileyonlinelibrary.com]

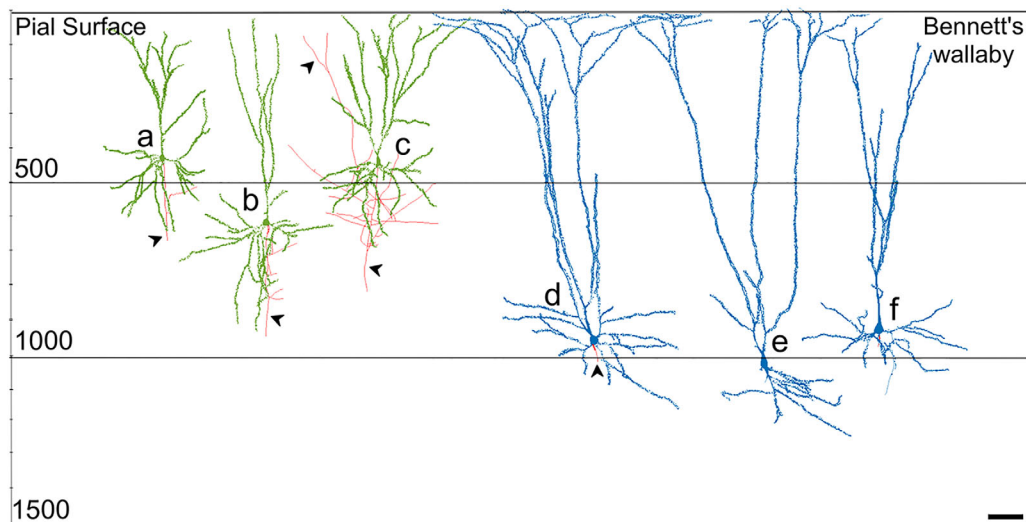


FIGURE 22 Neurolucida tracings of neurons in the primary motor cortex of the Bennett's wallaby indicating relative soma depth from the pial surface (in μm): superficial pyramidal neurons (a–c) and deep pyramidal neurons (d–f). Note the bifurcating apical dendrites in the deep pyramidal neurons that ascend nearly parallel to each other toward the pial surface. No gigantopyramidal neurons were distinguishable in the Golgi-impregnation for this species. Arrowheads indicate axons from neurons (a), (b), (c), and (d). Scale bar = 100 μm [Color figure can be viewed at wileyonlinelibrary.com]

dense basilar skirt appear to be narrow-spiking neurons characteristic of fast pyramidal tract neurons (fig. 9 in Chen, Zhang, Hu, & Wu, 1996).

The current stereological findings are the first in carnivores, and are in general agreement with the morphological results. Briefly, the soma volumes of gigantopyramidal neurons were greater than those of the corresponding layer V pyramidal neurons; however, this size differential was markedly less in primates (~ 2 times) and non-P carnivores (~ 4 times) than it was in the *Panthera* species (~ 12 times). Additionally, the somata of layer V pyramidal neurons in the *Panthera* species were no larger than one would expect when compared to non-P carnivores. However, the somata of *Panthera* gigantopyramidal neurons were larger than the corresponding somata of non-P carnivores (by 2.0 times by area, and 3.0 times by volume), and even larger than the corresponding somata in primates (by 4.6 times by area, and 12.1 times by volume). Thus, within the non-P carnivores, a distinct and statistically significant allometric relationship between brain or body mass and the size of layer V pyramidal neurons was observed, such that a doubling in brain mass led to a 1.10 times increase in layer V pyramidal neuron soma volume, and a doubling in body mass led to a 1.07 times increase in layer V pyramidal neuron soma volume. Notably, however, there was no significant association between brain or body mass and gigantopyramidal neuron size in these species. This is consistent with earlier predictions of a greater increase in average neuronal size in non-primate mammals associated with increased size of the cerebral cortex and lower cortical neuronal densities in non-primate mammals with similar cerebral cortical volumes to primates (Herculano-Houzel, Manger, & Kaas, 2014; Jardim-Messeder et al., 2017).

4.3 | Perissodactyls and artiodactyls

Although gigantopyramidal neurons in the African wild dog resembled those in the mountain zebra, and although molecular evidence suggests

a close phylogenetic relationship between perissodactyls and carnivores (Arnason et al., 2002; Nery, González, Hoffmann, & Opazo, 2012), there appeared to be overall greater morphological similarity between neurons in perissodactyls and artiodactyls in the present study. Therefore, these two orders are considered together here. The earliest mention of gigantopyramidal neurons in these orders was in sheep by Lewis (1878), who found that these neurons were arranged in “nests” similar to what had been observed by Betz (1874) in dogs, but that they also tended to be smaller than those found in humans or the domestic cat. Large layer V pyramidal neurons have been observed in the putative motor cortices in several cetacean species (Hof, Chanis, & Marino, 2005; Hof & Van der Gucht, 2007; Kesarev & Malofeeva, 1969), with potential morphological similarities to gigantopyramidal neurons documented in the humpback whale (Butti et al., 2015), but it remains unclear if these actually constitute gigantopyramidal motor neurons. In the present morphological analysis, gigantopyramidal neuron soma size in artiodactyls and perissodactyls was relatively small, perhaps because the corticospinal tract in ungulates does not project farther than the cervical level of the spinal cord (Badlangana et al., 2007; Nieuwenhuys, ten Donkelaar, & Nicholson, 1998). It should be noted, however, that the spinal cord in the giraffe remains very long (up to 2.6 m; Badlangana et al., 2007), which may explain why giraffe gigantopyramidal neurons in the present study have a dendritic volume ($94,701 \mu\text{m}^3$) considerably larger than any of the other ungulates examined (Jacobs, Harland, et al., 2015).

In the present study, a distinguishing characteristic for all species in these orders was the tendency for gigantopyramidal neurons to exhibit bifurcating apical branches. This forking apical geometry is consistent with that illustrated by Barasa (1960) in the domestic pig, cow, and horse, and has been previously documented in some cetartiodactyls (dolphin: Ferrer & Perera, 1988; giraffe: Jacobs, Harland, et al., 2015; pygmy hippopotamus: Butti et al., 2014; humpback whale: Butti et al., 2015). In

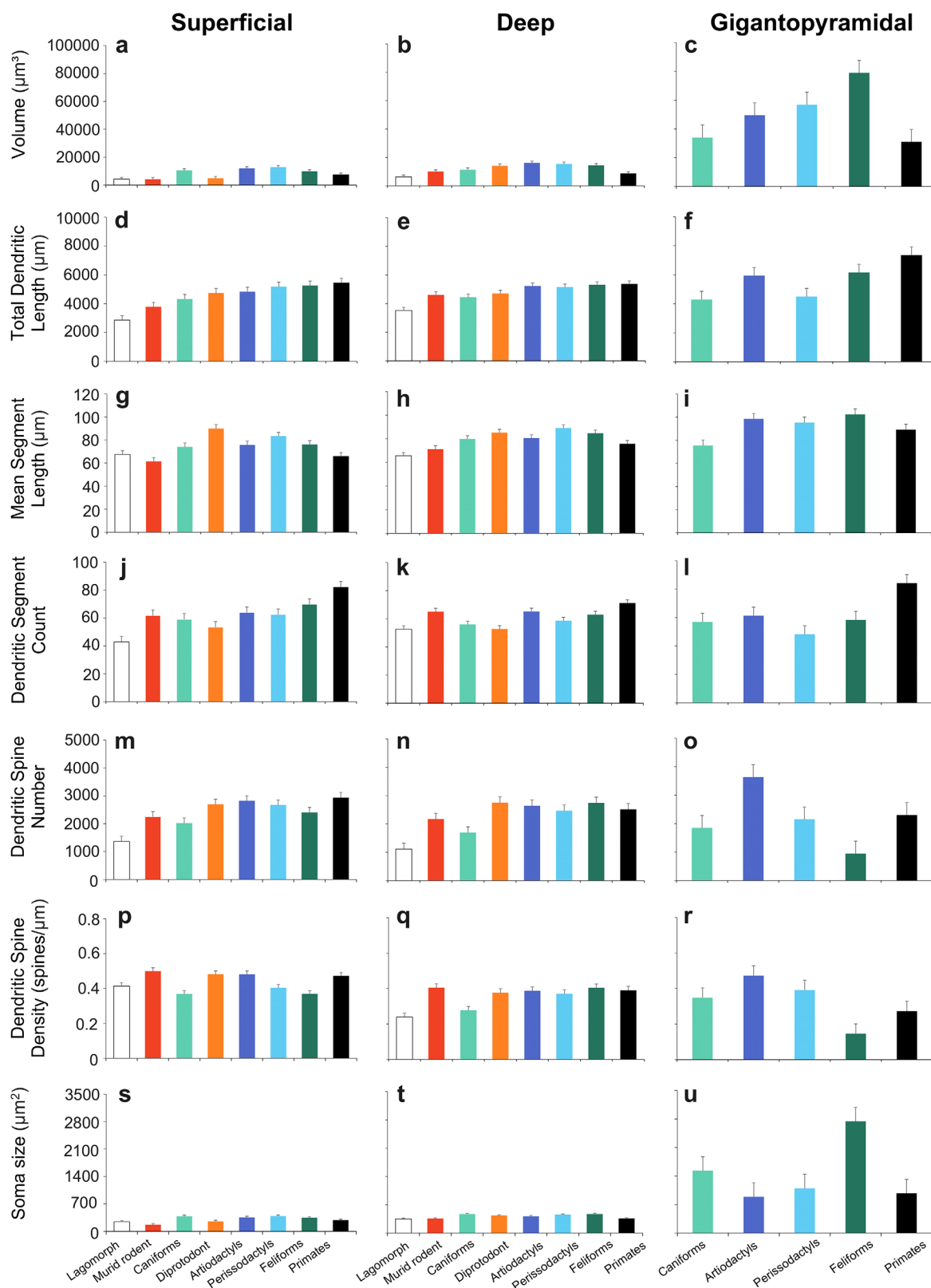


FIGURE 23 Bar graphs illustrating the average values of seven dependent measures (Vol, TDL, MSL, DSC, DSN, DSD, and soma size) for each taxon in the present morphological analysis. Columns are arranged by neuron type: superficial pyramidal neurons (a, d, g, j, m, p, s), deep pyramidal neurons (b, e, h, k, n, q, t), and gigantopyramidal neurons (c, f, i, l, o, r, u). Taxa for all graphs are arranged from left to right on the abscissa in a fixed order based on the ascending TDL values of superficial pyramidal neurons in graph (d). Note that gigantopyramidal neurons were not distinguishable in the diprotodont, lagomorph, or rodent. Error bars = SEM [Color figure can be viewed at wileyonlinelibrary.com]

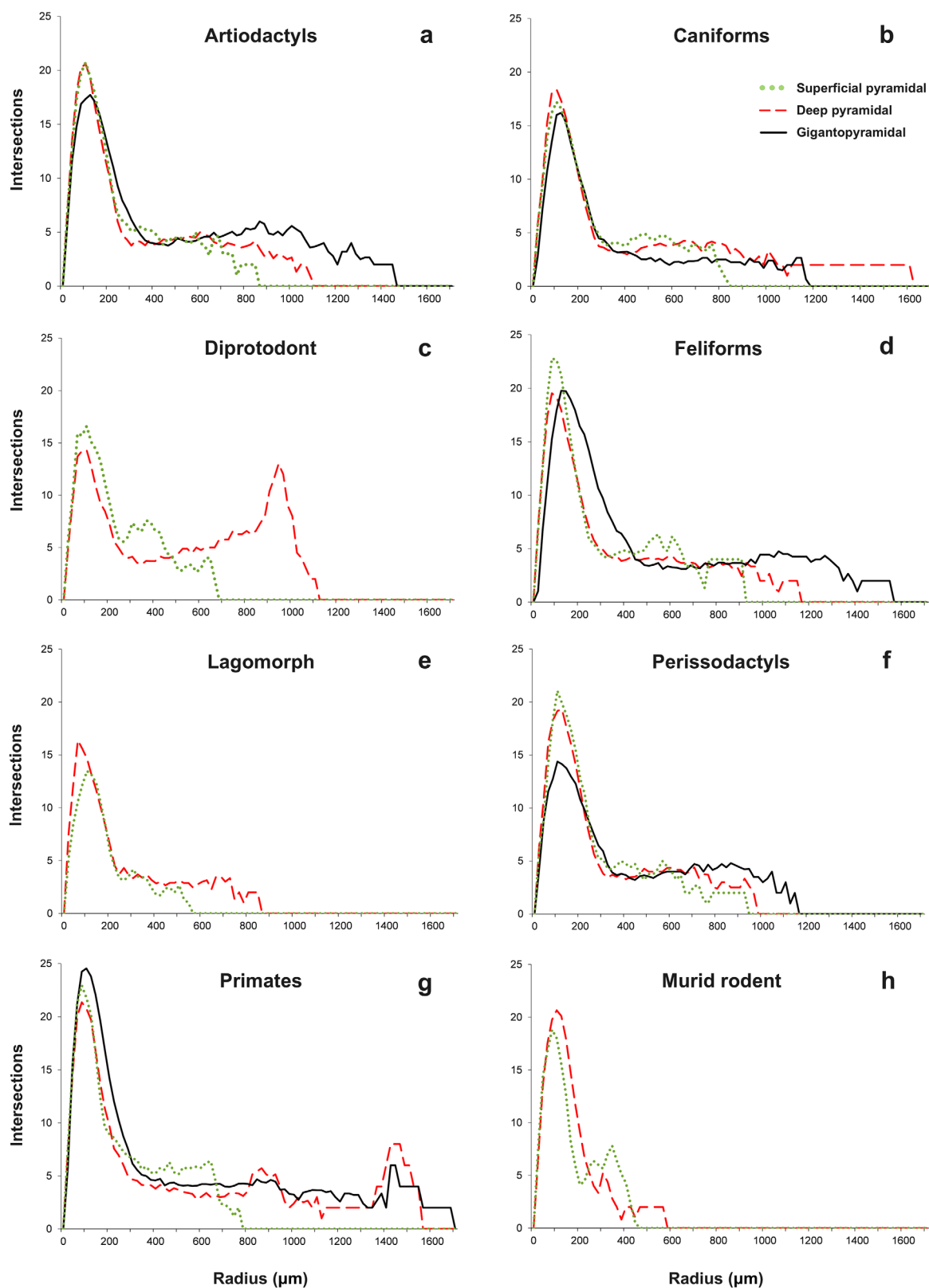


FIGURE 24 Sholl analyses of the three neuron types (superficial pyramidal, deep pyramidal, and gigantopyramidal) arranged by taxon, indicating relative dendritic complexity. The number of dendritic intersections (ordinate) were quantified at 20- μm intervals from the soma (abscissa) using concentric rings. Note that gigantopyramidal neurons were not distinguished in the diprotodont (wallaby), lagomorph (rabbit), or rodent (rat). For all neuron types, peak number of intersections occurred between 100 and 200 μm . Deep pyramidal and gigantopyramidal neuron dendrites tended to be similar to each other, with gigantopyramidal neuronal dendrites usually extending further from the soma [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 6 Confusion matrix of actual versus predicted taxonomic group for superficial pyramidal neurons with complete morphological data

Actual/predicted	Artiodactyls	Caniforms	Feliforms	Perissodactyls	Primates	Total	Correct	Incorrect
Artiodactyls	43	0	0	0	0	43	43	0
Caniforms	0	49	0	0	0	49	49	0
Feliforms	0	0	41	0	0	41	41	0
Perissodactyls	0	0	0	15	0	15	15	0
Primates	0	0	0	0	39	39	39	0
Diprotodont	0	2	1	5	1	9	0	9
Rodent	0	6	0	0	4	10	0	10
Lagomorph	0	12	0	0	2	14	0	14
Total	100%	71%	98%	75%	85%	85%	187	33

both orders, basilar dendrites tended to either descend off the base of the soma or assume the asymmetric arrangement of horizontal, circumferential dendrites observed in humans (Scheibel & Scheibel, 1978a).

4.4 | Primates

Since originally being discovered in humans (Betz, 1874), gigantopyramidal neurons have been extensively examined in primates with regard to soma size and cell counts (Brodman, 1909; Lassek, 1940), tract-tracing (Catsman-Berrevoets & Kuypers, 1976), descriptive morphology (Braak & Braak, 1976; Scheibel & Scheibel, 1978a; Scheibel et al., 1974), and clinicopathology (Hammer et al., 1979; Sasaki & Iwata, 2001; Scheibel et al., 1977; Tigges, 1992; Tigges, Herndon, & Peters, 1992; Tsuchiya et al., 2000, 2002; Udaka, Kameyama, & Tomonaga, 1986). Although gigantopyramidal neurons have also been observed in the lemur (Mott & Halliburton, 1908), as well as the baboon and tamarin (Sherwood et al., 2003), the present investigation is the first to examine dendritic architecture in these primates. Gigantopyramidal neuronal morphology was remarkably uniform across the four primate species examined, and consistent with previous observations (Barasa, 1960; Braak & Braak, 1976; Conel, 1939–1967; Meyer, 1987; Scheibel et al., 1974; Scheibel & Scheibel, 1978a). Larger layer V pyramidal neurons with descending taproots similar to the gigantopyramidal neurons observed here have also been identified as fast pyramidal tract neurons in the monkey (species unspecified; Hamada, Sakai, & Kubota, 1981). They have also been shown in macaque monkeys to express

parvalbumin-like immunoreactivity (Preuss & Kaas, 1996) and have stained densely for cytochrome oxidase (Matelli, Luppino, & Rizzolatti, 1985), both of which indicate potentially high firing rates and metabolic activity. Finally, current evidence suggests that large layer V pyramidal neurons, including gigantopyramidal neurons, in macaque M1 express the fast potassium channel, Kv3.1b, which has been associated with high-frequency, short duration action potentials typical of corticofugal pyramidal neurons (Ichinohe et al. 2004; Soares et al., 2017).

As in the morphological investigation, the stereological analysis revealed that the soma size of both layer V pyramidal and gigantopyramidal neurons in the primates were smaller than those observed in the carnivores. The soma volume of gigantopyramidal neurons was also greater than that of the corresponding layer V pyramidal neurons, consistent with observations of relative large soma volumes of Meynert and Betz cells compared to neighboring pyramidal neurons across a broad range of primates (Sherwood et al., 2003). The stereological measures, particularly soma volume, indicated that there was no allometric increase in the size of layer V pyramidal neurons in primates with brain or body mass. This finding parallels predictions about average neuronal sizes in the cerebral cortex of primates of varying size, where increases in average neuronal size only occurred with a very shallow negative allometry (Herculano-Houzel et al., 2014). The fact that the layer V pyramidal neuron somata of primates in the present study were smaller than those of non-P carnivores of similar cortical volumes is also potentially supportive of the predictions made through isotropic fractionator analysis due to the higher neuronal density

TABLE 7 Confusion matrix of actual versus predicted taxonomic group for gigantopyramidal neurons with complete morphological data

Actual/predicted	Artiodactyls	Feliforms	Perissodactyls	Primates	Caniforms	Total	Correct	Incorrect
Artiodactyls	22	0	0	0	7	29	22	7
Feliforms	1	16	0	0	0	17	16	1
Perissodactyls	0	0	17	0	0	17	17	0
Primates	0	0	0	45	0	45	45	0
Caniforms	2	1	7	3	1	14	1	13
Total	88%	94%	71%	94%	13%	83%	101	21

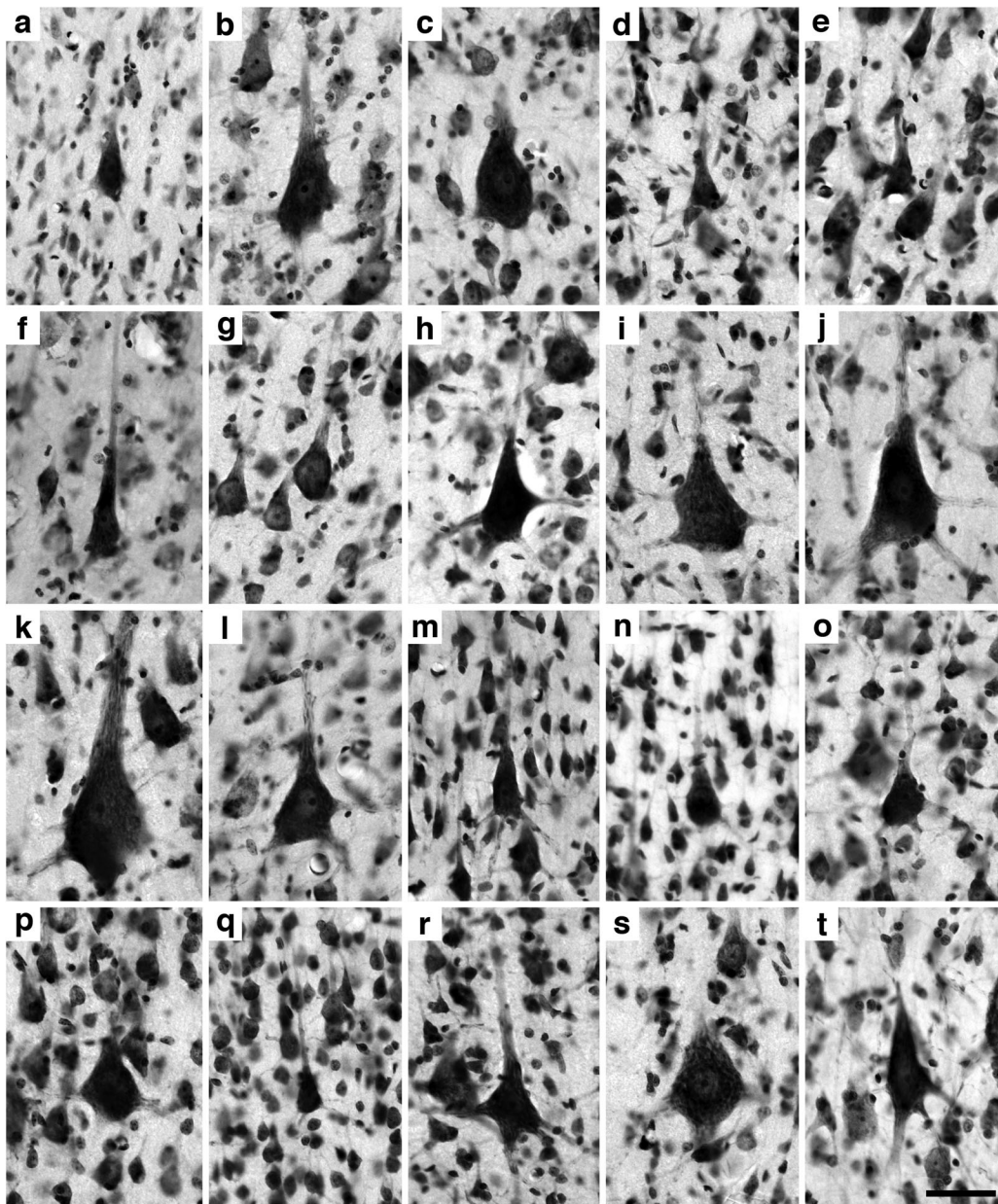


FIGURE 25 High power photomicrographs of Nissl stained gigantopyramidal cells from layer V of the primary motor cortex in 11 carnivore species (a–k) and 9 primate species (l–t). (a) *Mustela putorius furo*, European polecat; (b) *Amblonyx cinereus*, Asian small-clawed otter; (c) *Procyon lotor*, raccoon; (d) *Pagophilus groenlandicus*, harp seal; (e) *Callorhinus ursinus*, northern fur seal; (f) *Lycaon pictus*, African wild dog*; (g) *Mungos mungo*, banded mongoose*; (h) *Caracal caracal*, caracal*; (i) *Panthera pardus orientalis*, Amur leopard; (j) *Panthera leo*, African lion*; (k) *Panthera tigris altaica*, Siberian tiger*; (l) *Cebuella pygmaea*, pygmy marmoset; (m) *Saguinus oedipus*, cotton-top tamarin; (n) *Leontopithecus rosalia*, golden lion tamarin*; (o) *Saimiri boliviensis*, black-capped squirrel monkey; (p) *Cercopithecus ascanius*, red-tailed monkey; (q) *Chlorocebus pygerythrus*, vervet monkey; (r) *Papio hamadryas*, hamadryas baboon; (s) *Papio ursinus*, chacma baboon*; (t) *Hylobates lar*, lar gibbon. Note the very large size of these pyramidal neurons in the representatives of the Genus *Panthera* (I–K). * denotes animals also in the morphological analysis. Scale bar in t = 50 μ m and applies to all panels

observed in primates compared to other mammals (Herculano-Houzel et al., 2014; Jardim-Messeder et al., 2017).

4.5 | Lagomorph, rodent, and diprotodont

Gigantopyramidal neurons were not morphologically identifiable in the mongoose, rabbit, rat, or wallaby. However, mongoose gigantopyramidal neurons were noted in the stereological analysis (Figure 25g),

although they remained quite small compared to those in other feliforms (Table 3). Historically, Brodmann (1909) observed that gigantopyramidal neurons in the rabbit were among the smallest in his specimens, and noted that it was difficult to distinguish them from other large layer V pyramidal neurons in species such as rodents and the wallaby (see also Ashwell, Zhang, & Marotte, 2005; Sherwood et al., 2009). Barasa's (1960) depictions of pyramidal neurons in the mouse, rat, and rabbit also suggest that gigantopyramidal neurons are,

TABLE 8 Summary of reduced major axis (RMA) regression and analyses of brain mass (BrM) or body mass (BoM) against the length (l), area (a), and volume (v) of layer V pyramidal (Vp) and gigantopyramidal (Vg) neuron somata in non-*Panthera* (non-P) carnivores and primates

Group	RMA regression	Slope	Intercept	r^2	P_{uncorr}
Non-P carnivores	BoM vs Vpl	0.040	0.777	.749	.005*
	BrM vs Vpl	0.055	0.844	.803	.003*
	BoM vs Vgl	0.088	0.768	.009	.823
	BrM vs Vgl	0.119	0.912	.037	.649
	BoM vs Vpa	0.093	2.025	.703	.019*
	BrM vs Vpa	0.133	2.164	.825	.005*
	BoM vs Vga	0.176	2.067	.007	.848
	BrM vs Vga	0.239	2.357	.034	.664
	BoM vs Vpv	0.129	3.012	.554	.034*
	BrM vs Vpv	0.176	3.225	.619	.021*
	BoM vs Vgv	0.259	3.049	.006	.855
	BrM vs Vgv	0.421	3.319	.019	.769
Primates	BoM vs Vpl	0.044	0.675	.088	.427
	BrM vs Vpl	0.061	0.726	.187	.245
	BoM vs Vgl	0.176	2.917	.087	.441
	BrM vs Vgl	0.244	3.120	.149	.303
	BoM vs Vpa	0.088	1.873	.116	.371
	BrM vs Vpa	0.122	1.975	.223	.199
	BoM vs Vga	0.093	2.069	.046	.578
	BrM vs Vga	0.129	2.177	.069	.496
	BoM vs Vpv	0.139	2.685	.123	.355
	BrM vs Vpv	0.194	2.847	.243	.178
	BoM vs Vgv	0.176	2.917	.087	.441
	BrM vs Vgv	0.244	3.120	.149	.304

* $p \leq .05$.

at best, difficult to identify in these species. Based on electrophysiological and morphological evidence, several laboratories have in fact concluded that gigantopyramidal neurons, as documented in primates and domestic cats (Deschênes et al., 1979; Hamada et al., 1981), are not clearly identifiable in the rat (Landry, Wilson, & Kitai, 1984; Tseng & Prince, 1993). Moreover, layer V pyramidal neurons in rat primary motor cortex appear to lack the fast potassium channel (Kv3.1b) observed in macaques (Soares et al., 2017). Thus, there appears to be little evidence that gigantopyramidal neurons can be unambiguously identified in these species.

4.6 | Gigantopyramidal neuron size and function speculations

Gigantopyramidal neurons in the present study were more variable in terms of somatodendritic volume than other pyramidal neurons. Historically, several potential explanations have been suggested to explain

species differences in gigantopyramidal neurons. As outlined and partially dismissed by Brodmann (1909), there is a hypothesized direct correlation between soma size and (a) body size, (b) axon length (Campbell, 1905; Lewis, 1878), and (c) the size of the dependent muscle (Jackson, 1890; Lassek, 1940). One could also hypothesize a direct relationship between soma size and (d) brain size, (e) the size of cerebral cortex, and (f) measures of digital dexterity. However, there appears to be no relation between digital dexterity and soma size in gigantopyramidal neurons (Sherwood et al., 2003), as is illustrated in the present sample because feliforms and caniforms, which rank low in digital dexterity (Heffner & Masterton, 1975), exhibited considerably larger gigantopyramidal neurons than primates, which rank high in digital dexterity. It does appear, however, that a positive correlation exists between soma size and body/cerebral cortex/brain size *within* mammalian lineages but not *across* them (Elston & Manger, 2014). Although each of these factors may influence the size of gigantopyramidal neurons, none of them appears sufficient on its own. Moreover, none of

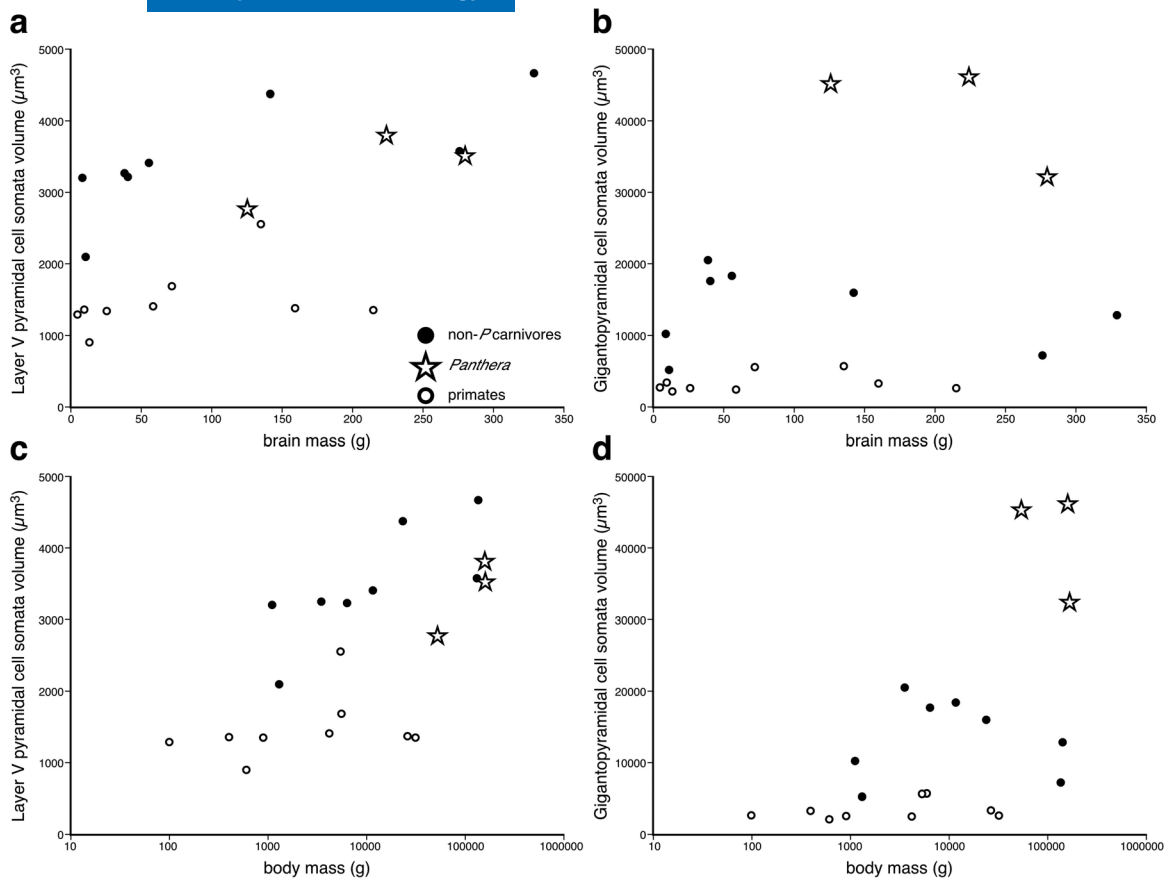


FIGURE 26 Scatterplots depicting the volume (μm^3) of layer V pyramidal neuron somata (a, c) and gigantopyramidal cell somata (b, d) for primates (open circles), non-*Panthera* carnivores (non-P, closed circles), and *Panthera* species (African lion, Amur leopard, Siberian tiger; open stars) in relation to brain mass (a, b) and body mass (c, d). Note that the somata of the layer V pyramidal and gigantopyramidal neurons are larger in carnivores than in primates and that the somata of gigantopyramidal neurons in the *Panthera* species are extremely large compared to all other species examined

these explanations address cross-species variations in the organization of descending multifunctional pathways (Kuypers, 1981), including the cortico-motoneuronal system (Illert, Lundberg, & Tanaka, 1976; for review, see Lemon, 2008; Lemon et al., 2004; Lemon & Griffiths, 2005). Unfortunately, a comprehensive account of such species-specific variation, which is now extending into molecular specializations across species (Soares et al., 2017), is beyond the scope of the present study.

Nevertheless, the cross-order comparative perspective presented here does facilitate speculation about one additional factor that may play a role in determining soma size in gigantopyramidal neurons, namely the muscle composition and accompanying (predatory) behavior in a given (sub)order. Most adult mammalian skeletal muscles appear to be heterogeneous in their fiber composition, with variation in fiber types across species and from muscle to muscle within a species (Hyatt, Roy, Rugg, & Talmadge, 2010; Kohn, Burroughs, Hartman, & Noakes, 2011; Kohn & Noakes, 2013; Toniolo et al., 2007). However, within a single motor unit, the muscle fiber composition is homogeneous (Goto et al., 2013; Schiaffino & Reggiani, 2011). Skeletal muscles are generally categorized according to their maximal contraction velocity (slow or fast), predominate bioenergetic pathways for adenosine triphosphate

production (ATP; oxidative or glycolytic) and, more precisely, on their expression of various myosin heavy chain (MHC) protein isoforms (Goto et al., 2013; Hyatt et al., 2010; Schiaffino & Reggiani, 2011). Excluding hybrid fiber types, there are generally four MHC isoforms of mammalian trunk and limb muscles (Type I, IIa, IIx, IIb; Kohn et al., 2011). Type I fibers are slow, fatigue-resistant, oxidative fibers that are innervated by smaller motoneurons. Type II fibers are fast, glycolytic fibers that are more powerful, lack fatigue resistance, and are innervated by larger motoneurons. As one progresses from type IIa to IIx and finally IIb isoforms, the fibers become faster, more reliant on glycolytic ATP production, more powerful, and more easily fatigued (Goto et al., 2013; Kohn et al., 2011; Schiaffino & Reggiani, 2011).

Skeletal muscles of larger mammals are mainly composed of type I, IIa, and IIx fibers, with little expression of the most powerful IIb type fibers in trunk and limb muscles (Goto et al., 2013; Kohn et al., 2011; Schiaffino & Reggiani, 2011; Toniolo et al., 2007). In humans, skeletal muscles of cyclists and runners have been found to be composed of mostly type I and IIa fibers, whereas sprinters and power athletes express mostly type IIa and IIx MHC isoforms (Kohn & Noakes, 2013; Schiaffino & Reggiani, 2011). In some caniforms (Goto et al., 2013; Toniolo et al., 2007), muscle fibers appear to be primarily type IIa fibers.

However, in feliforms (e.g., African lion, caracal: Kohn et al., 2011; Kohn & Noakes, 2013; cheetah, *Acinonyx jubatus*: Goto et al., 2013; Hyatt et al., 2010), including the domestic cat, muscle fibers typically contain high percentages of type IIx fibers, with fewer type I fibers. Even the same MHC isoforms may differ across species. For example, caracal and lion type IIx fibers appear to produce substantially greater force and more power than the human hybrid IIax fibers, which may partially explain the speed, strength, and jumping abilities in these felids (Kohn & Noakes, 2013). It thus appears that muscle fiber composition is closely aligned with the physical activity profiles in different animals (Curry, Hohl, Noakes, & Kohn, 2012).

As such, the current findings of larger gigantopyramidal neurons in feliforms may relate to differences in muscle fiber composition and motor control systems. For predatory animals such as felids, motor control is a high priority, specialized system. Some hindlimb muscles examined in the cheetah, for example, appear to possess the fastest MHC isoform (IIb), which potentially contributes to its propulsive power and sprinting ability (Goto et al., 2013; Hyatt et al., 2010). In contrast, some hindlimb muscles in the Siberian tiger contain a high combined percentage of MHC I and IIa fibers, facilitating its ability to cover long distances in its daily territory patrols (Hyatt et al., 2010). To maximize energy returns, tigers and many other species in the *Panthera* genus have an optimal foraging strategy that favors preying on species of similar body mass to themselves (Funston, Mills, Biggs, & Richardson, 1998; Haywood, Jędrzejewski, & Jędrzejewska, 2012) which, in turn, requires specialized hunting techniques. Many panthers hunt by stalking and pouncing on prey, which demands a substantial burst of energy, considerable strength, and the quick recruitment/coordination of many muscle groups. In modulating downstream neuromuscular output, gigantopyramidal neurons in the motor cortex exhibit several functional characteristics that may make them particularly important for feliforms: (1) they appear to be able to synchronize their activity because of dendritic bundling (Meyer, 1987; Scheibel et al., 1974) and clustering (Groos, Ewing, Carter, & Coulter, 1978), particularly in the hand-forepaw region (Rivara et al. 2003); (2) they exhibit increased conduction velocities because of their large axonal diameter (Evarts, 1965; Firmin et al., 2014; Sakai & Woody, 1988), and are associated, at least indirectly, with increased striated muscle contraction velocity and power (Buller, Eccles, & Eccles, 1960; Kohn & Noakes, 2013; Salmons & Vrbová, 1969); and finally, (3) they appear to initiate incipient, patterned downstream motor activity quickly (Lundberg & Voorhoeve, 1962; Takahashi, 1965). These functional characteristics of gigantopyramidal neurons seem consistent with, if not a prerequisite for, the unusually short reaction times in feliforms (Kohn et al. 2011; Kohn & Noakes, 2013). Thus, the exceptionally large gigantopyramidal neurons observed here for feliforms, specifically for *Panthera*, may have co-evolved with their specialized cortico-motoneuronal-musculoskeletal system (Gonyea, 1976; Hyatt et al., 2010), which contributes to the behavioral repertoires of these species. A comprehensive account of gigantopyramidal neuron size in the motor cortex should therefore also consider muscle fiber composition profiles in addition to factors such as brain size and body size.

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CONFLICT OF INTEREST

The authors have no conflicts of interest.

AUTHOR CONTRIBUTIONS

All authors had full access to all data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis. Study concept and design: PRM, CCS, PRH, BJ, MS. Collection of and qualitative analysis of data: MEG, NBSS, MET, PRH, PRM, MAS, AB, HKC. Statistical analysis and interpretation: MS, BJ, MSS, MAS. Procurement, preparation, and fixation of tissue: AHL, AB, BW, PRM, CCS, LLD, MFB, MAS, TW. Obtained funding: PRM, PRH, CCS, BJ. Drafting of the manuscript: BJ, MS, NBSS, MEG, MET, AJB, PRM, PRH. Photomicrography and preparation of figures/tables: PRM, BJ, MET, MEG, NBSS, TAT, MS, CCS. Critical revision of the manuscript for important intellectual content: PRM, PRH, CCS, MEG, NBSS, MET, TAT, MS, MAR, BJ. Study supervision: BJ, PRH, PRM, CCS.

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